Calculus 1 Theory

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Introduction

This document is a personal collection of the arguments and formulas I studied in Calculus 1. It is not a textbook and it does not aim at completeness. The goal is to keep in one place the theoretical material I want to be able to recall: definitions, main theorems, typical proof ideas, and the identities that come up when solving problems. The notes follow the usual order of a first course in calculus of one real variable: the real line and limits, continuity, derivatives, applications of differentiation, integrals, and the fundamental theorems that connect them. Each later section should state the result clearly and record the argument at the level I used when studying it, without extra commentary. I will use these pages as a reference while practising exercises and as a compact review before exams. When a formula or a theorem has a standard name, that name is kept so that it matches lectures and textbooks.

1 Formalities: mathematical language

Mathematics is written with a small number of precise words. The four that appear everywhere later are *predicate*, *statement*, *universal implication* and *theorem*.

1.1 Predicate

Definition 1 (Predicate)**.** A *predicate* $P(x)$, $P(x,y)$, $\dots$ is an assertion that contains one or more free variables. On its own it is neither true nor false: it becomes true or false only after the variables are specified, or after they are bound by a quantifier.

Examples:

$$\begin{aligned} P(x)&: \quad x>0,\\ Q(x,y)&: \quad x^2+y^2=1,\\ R(n)&: \quad n \text{ is even.} \end{aligned}$$

$P(3)$ is true, $P(-1)$ is false. The expression $P(x)$ itself is not a statement. The usual quantifiers turn a predicate into a statement:

$$\forall x \in A,\ P(x) \qquad \text{and} \qquad \exists x \in A,\ P(x).$$

The first is true when $P(x)$ holds for every $x \in A$; the second when it holds for at least one $x \in A$.

1.2 Statement

Definition 2 (Statement)**.** A *statement* is a declarative sentence that is either true or false, and not both. It has no free variables.

Examples of statements: $2+2=4$ (true); $\sqrt{2}$ is rational (false); $\forall x \in \mathbb{R},\ x^2 \geq 0$ (true).

1.3 Universal implication

Most statements in calculus are not a single implication $P \Rightarrow Q$, but an implication that is required to hold for every object of a given type.

Definition 3 (Universal implication)**.** A *universal implication* is a statement of the form

$$\forall x \in A, \quad P(x) \Rightarrow Q(x).$$

$P(x)$ is the *hypothesis* and $Q(x)$ is the *conclusion.* It is read: for every $x \in A$, if $P(x)$ holds then $Q(x)$ holds.

The universal implication is true when every $x \in A$ that satisfies the hypothesis also satisfies the conclusion. It is false if and only if there exists a *counterexample*: an element $x_0 \in A$ such that $P(x_0)$ is true and $Q(x_0)$ is false. To prove a universal implication one typically takes an arbitrary $x \in A$ with $P(x)$ and deduces $Q(x)$. To disprove it, it is enough to exhibit one counterexample. The converse is

$$\forall x \in A, \quad Q(x) \Rightarrow P(x).$$

It is a different statement; it may be true or false independently of the original implication. If both hold, one writes

$$\forall x \in A, \quad P(x) \Leftrightarrow Q(x).$$

Then P is a *necessary and sufficient* condition for Q.

1.4 Theorem

Definition 4 (Theorem)**.** A *theorem* is a statement that has been proved to be true. In these notes a theorem is almost always a universal implication, or a chain of such implications.

The same kind of true statement may be labelled in a weaker or stronger way according to its role:

- a *lemma* is a theorem used mainly as a step toward another result;
- a *proposition* is a theorem of intermediate importance;
- a *corollary* is a theorem that follows in a short way from a previous one.

A *definition* is not a theorem: it introduces a name for an object or a property; it is neither proved nor disproved. The standard pattern of a theorem in Calculus 1 is therefore:

> **Theorem.** $\forall x \in A$, if $P(x)$ then $Q(x)$.

followed by a *proof*: a finite sequence of allowed deductions that starts from the hypothesis and ends at the conclusion.

1.5 Law of contrapositives

Definition 5 (Law of contrapositives)**.** The implication $P \Rightarrow Q$ is equivalent to its *contrapositive* $\sim Q \Rightarrow \sim P$:

$$(P \Rightarrow Q) \quad \Leftrightarrow \quad (\sim Q \Rightarrow \sim P).$$

The two statements are true together or false together. In particular they are not the converse $Q \Rightarrow P$.

For a universal implication the same law reads

$$(\forall x \in A,\ P(x) \Rightarrow Q(x)) \quad \Leftrightarrow \quad (\forall x \in A,\ \sim Q(x) \Rightarrow \sim P(x)).$$

Therefore, to prove that $P(x)$ implies $Q(x)$ for every $x \in A$, it is enough to prove that whenever $Q(x)$ fails, $P(x)$ fails as well: take an arbitrary $x \in A$ with $\sim Q(x)$ and deduce $\sim P(x)$.

1.6 Proof by contradiction

Definition 6 (Proof by contradiction)**.** To prove a statement R *by contradiction*, assume $\sim R$ and deduce from that assumption a false statement (a contradiction, typically of the form $S \wedge \sim S$). Then $\sim R$ cannot hold, so R is true.

When R is a universal implication $\forall x \in A,\ P(x) \Rightarrow Q(x)$, a proof by contradiction proceeds as follows. Suppose the implication is false: there exists $x_0 \in A$ such that $P(x_0)$ and $\sim Q(x_0)$. From these two assumptions one derives a contradiction. Hence no such x_0 exists, and the implication holds. This is not the same argument as the law of contrapositives. The contrapositive proves $\sim Q \Rightarrow \sim P$ directly. A proof by contradiction assumes the negation of the claim one wants, and reaches an impossibility.

2 Geometric progression

A *geometric progression* (GP) is a sequence of real numbers

$$(a_n)_{n\geq 0}$$

such that every term is obtained by multiplying the previous one by a fixed real number q, called the *ratio* of the progression.

Definition 7 (Geometric progression)**.** A sequence (a_n) is a geometric progression of ratio q if

$$a_{n+1} = q\, a_n \qquad \text{for all } n \geq 0.$$

Starting from the initial term a_0, the general term is

$$a_n = a_0\, q^n.$$

Geometric sum

The basic geometric sum is

$$\sum_{k=0}^{n} q^k = \frac{1-q^{n+1}}{1-q} \qquad (q \neq 1).$$

When $q = 1$, every term equals 1 and

$$\sum_{k=0}^{n} 1 = n+1.$$

From this, the sum of the first $n+1$ terms of a geometric progression is

$$S_n = \sum_{k=0}^{n} a_k = a_0 \sum_{k=0}^{n} q^k = a_0\, \frac{1-q^{n+1}}{1-q} \qquad (q \neq 1).$$

If $q = 1$, the progression is constant and

$$S_n = (n+1)a_0.$$

Infinite geometric series

When $|q| < 1$, the geometric progression has a convergent infinite sum:

$$\sum_{n=0}^{\infty} a_n = \frac{a_0}{1-q}.$$

If $|q| \geq 1$, the infinite series diverges.

Examples

$$\begin{aligned} a_n &= 3 \cdot 2^n && \text{GP with } a_0 = 3,\ q = 2 \\ a_n &= 5\left(\tfrac{1}{2}\right)^n && \text{GP with } a_0 = 5,\ q = \tfrac{1}{2} \end{aligned}$$

3 Binomial coefficients and Newton's formula

Binomial coefficients

For every pair of integers $n \geq 0$ and $0 \leq k \leq n$, the *binomial coefficient* $\binom{n}{k}$ counts the number of ways to choose k elements from a set of n elements. It is defined by

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

The binomial coefficients satisfy the symmetry identity

$$\binom{n}{k} = \binom{n}{n-k},$$

and the recurrence relation

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1},$$

which is the rule that generates Pascal's triangle.

Newton's binomial formula

For every real numbers x, y and every integer $n \geq 0$, the expansion of $(x+y)^n$ is given by the *binomial formula*:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Each term corresponds to choosing k factors y and $n-k$ factors x in the product $(x+y)(x+y)\cdots(x+y)$ repeated n times.

Special cases

Setting $x = 1$ gives the identity

$$(1+y)^n = \sum_{k=0}^{n} \binom{n}{k} y^k.$$

Setting $y = 1$ gives

$$(x+1)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k}.$$

Setting $x = y = 1$ yields the classical sum of the n-th row of Pascal's triangle:

$$2^n = \sum_{k=0}^{n} \binom{n}{k}.$$

4 Fields and ordered fields

Fields

A *field* is a set F equipped with two binary operations, addition and multiplication, satisfying the axioms listed below. The elements of F are called *numbers*. The operations are written $x + y$ and xy (or $x \cdot y$).

R1: Axioms for addition

The addition operation $+$ satisfies the following properties for all $x, y, z \in F$:

- **(R1.1) Commutativity:** $x + y = y + x$.
- **(R1.2) Associativity:** $(x + y) + z = x + (y + z)$.
- **(R1.3) Additive identity:** there exists an element $0 \in F$ such that $x + 0 = x$ for all $x \in F$.
- **(R1.4) Additive inverse:** for every $x \in F$ there exists an element $-x \in F$ such that $x + (-x) = 0$.

The element 0 is called the *additive identity*, and $-x$ is the *additive inverse* of x.

R2: Axioms for multiplication

The multiplication operation $\cdot$ satisfies the following properties for all $x, y, z \in F$:

- **(R2.1) Commutativity:** $xy = yx$.
- **(R2.2) Associativity:** $(xy)z = x(yz)$.
- **(R2.3) Multiplicative identity:** there exists an element $1 \in F$, $1 \neq 0$, such that $x \cdot 1 = x$ for all $x \in F$.
- **(R2.4) Multiplicative inverse:** for every $x \in F$ with $x \neq 0$ there exists an element $x^{-1} \in F$ such that $x \cdot x^{-1} = 1$.

The element 1 is called the *multiplicative identity*, and x^{-1} is the *multiplicative inverse* of x.

Distributive law

The two operations are connected by the distributive property:

$$x(y + z) = xy + xz \qquad \text{for all } x, y, z \in F.$$

A set F satisfying R1, R2, and the distributive law is called a *field*.

R3: Order relation

An *order relation* on F is a binary relation $\leq$ satisfying, for all $x, y, z \in F$:

- **(R3.1) Totality:** exactly one of the following holds: $x < y$, $x = y$, or $x > y$.
- **(R3.2) Transitivity:** if $x \leq y$ and $y \leq z$, then $x \leq z$.
- **(R3.3) Compatibility with addition:** if $x \leq y$, then $x + z \leq y + z$.
- **(R3.4) Compatibility with multiplication:** if $0 \leq x$ and $0 \leq y$, then $0 \leq xy$.

The relation $\leq$ is called an *order* on F.

Ordered fields

A field F equipped with an order relation $\leq$ satisfying R3 is called an *ordered field*. In an ordered field one writes $x > 0$ to mean $0 < x$, and $x < 0$ to mean $x < 0$. The standard examples of ordered fields are $\mathbb{Q}$ and $\mathbb{R}$.

5 Supremum and Infimum

Let $A \subseteq \mathbb{R}$ be a nonempty set. Since $\mathbb{R}$ is an ordered field ("The relation $\leq$ is called an order on F." and "The standard examples of ordered fields are $\mathbb{Q}$ and $\mathbb{R}$."), we can compare its elements and define upper and lower bounds.

Upper bounds and supremum

Definition 8 (Upper bound)**.** A real number M is an *upper bound* of A if

$$x \leq M \qquad \text{for all } x \in A.$$

Definition 9 (Supremum)**.** The *supremum* (least upper bound) of A, denoted $\sup A$, is the smallest real number that is an upper bound of A. A real number s is $\sup A$ if and only if:

- every $x \in A$ satisfies $x \leq s$;
- for every $\varepsilon > 0$ there exists $x \in A$ such that $x > s - \varepsilon$.

The second condition means that no number smaller than s can be an upper bound.

Lower bounds and infimum

Definition 10 (Lower bound)**.** A real number m is a *lower bound* of A if

$$m \leq x \qquad \text{for all } x \in A.$$

Definition 11 (Infimum)**.** The *infimum* (greatest lower bound) of A, denoted $\inf A$, is the largest real number that is a lower bound of A. A real number t is $\inf A$ if and only if:

- every $x \in A$ satisfies $t \leq x$;
- for every $\varepsilon > 0$ there exists $x \in A$ such that $x < t + \varepsilon$.

Existence in $\mathbb{R}$

Every nonempty set that is bounded above has a supremum, and every nonempty set that is bounded below has an infimum. This property is the *completeness* of the real numbers.

Examples

- For $A = (0, 1)$, $\sup A = 1$ and $\inf A = 0$.
- For $A = \{1/n : n \in \mathbb{N}\}$, $\sup A = 1$ and $\inf A = 0$.
- For a finite set, $\sup A$ is its maximum and $\inf A$ is its minimum.

R4: Supremum property

Let X be an ordered field. We say that X satisfies the *supremum property* if: **(R4) Supremum property (continuity axiom):** every nonempty subset $E \subseteq X$ that is a proper subset of X and is bounded above admits a supremum in X. This property is also called the *continuity axiom* of the ordered field. If X satisfies (R4), then every nonempty subset $E \subseteq X$ that is bounded below admits an infimum in X. Indeed, one may apply (R4) to the set of all lower bounds of E, or equivalently to the set $-E = \{-x : x \in E\}$, using the correspondence

$$\inf E = -\sup(-E).$$

The ordered field $\mathbb{Q}$ does *not* satisfy (R4): there exist nonempty subsets of $\mathbb{Q}$ that are bounded above but do not have a supremum in $\mathbb{Q}$. A classical example is

$$E = \{\, q \in \mathbb{Q} : q^2 < 2 \,\},$$

which is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$, since $\sqrt{2} \notin \mathbb{Q}$. The ordered field $\mathbb{R}$ *does* satisfy (R4): every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$, and every nonempty subset that is bounded below has an infimum in $\mathbb{R}$.

Axiomatic definition of $\mathbb{R}$

A *real number system* $\mathbb{R}$ can be defined axiomatically as a set equipped with two operations $+$ and $\cdot$ and an order relation $\leq$ such that:

- $\mathbb{R}$ is a field: it satisfies R1 (axioms for addition), R2 (axioms for multiplication), and the distributive law;
- $\mathbb{R}$ is an ordered field: it is equipped with an order relation $\leq$ satisfying R3 (order axioms);
- $\mathbb{R}$ satisfies R4: the supremum property (continuity axiom).

Thus $\mathbb{R}$ is an *ordered field with the supremum property*, and this completeness axiom distinguishes $\mathbb{R}$ from $\mathbb{Q}$.

6 Triangle inequality

In the ordered field $\mathbb{R}$, the absolute value of a real number x is defined by

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

The absolute value measures the distance of x from 0 on the real line. The fundamental estimate relating absolute values is the *triangle inequality*.

Theorem 1 (Triangle inequality)**.** *For all real numbers $x, y \in \mathbb{R}$,*

$$|x + y| \leq |x| + |y|.$$

This inequality expresses that the distance from 0 to $x + y$ is never greater than the sum of the distances from 0 to x and from 0 to y.

Reverse triangle inequality

A direct consequence of the triangle inequality is the *reverse triangle inequality*:

$$\big|\,|x| - |y|\,\big| \leq |x - y|.$$

It states that the difference between the lengths $|x|$ and $|y|$ cannot exceed the length of their difference.

Geometric interpretation

On the real line, $|x-y|$ represents the distance between x and y. The triangle inequality corresponds to the fact that in any triangle, the length of one side is at most the sum of the lengths of the other two.

Examples

- $|3 + (-5)| = |-2| = 2 \leq |3| + |-5| = 8$.
- $|1.2 + 4.7| = 5.9 \leq 1.2 + 4.7 = 5.9$ (equality case).
- Reverse inequality: $\big|\,|7| - |2|\,\big| = 5 \leq |7 - 2| = 5$.

7 Principle of mathematical induction

Many statements in calculus and discrete mathematics concern all integers greater than or equal to a fixed starting point. The *principle of mathematical induction* is the fundamental method used to prove such statements.

Definition 12 (Principle of induction)**.** Let $P(n)$ be a statement depending on an integer $n \geq n_0$. We say that $P(n)$ holds for all integers $n \geq n_0$ if the following two conditions are satisfied:

- **(Base step)** The statement $P(n_0)$ is true.
- **(Inductive step)** For every integer $k \geq n_0$, if $P(k)$ is true, then $P(k+1)$ is true.

Under these two assumptions, $P(n)$ is true for all integers $n \geq n_0$.

The base step verifies the statement at the starting point. The inductive step shows that the truth of $P(k)$ forces the truth of $P(k+1)$, creating a chain of implications:

$$P(n_0) \Rightarrow P(n_0 + 1) \Rightarrow P(n_0 + 2) \Rightarrow \cdots$$

Strong induction

A useful variant is *strong induction*. It states that $P(n)$ holds for all $n \geq n_0$ if:

- $P(n_0)$ is true;
- for every $k \geq n_0$, if $P(n_0), P(n_0 + 1), \ldots, P(k)$ are all true, then $P(k+1)$ is true.

Strong induction is equivalent to ordinary induction, but often simplifies proofs where the inductive step depends on several previous cases.

Example

To prove that

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

for all integers $n \geq 1$:

- Base step: for $n = 1$, both sides equal 1.
- Inductive step: assume the formula holds for $n = k$. Then

  $$\sum_{i=1}^{k+1} i = \left(\sum_{i=1}^{k} i\right) + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2},$$

  which is the formula for $n = k + 1$.

Thus the identity holds for all $n \geq 1$ by induction.

8 Complex numbers

Real numbers are not sufficient to solve all algebraic equations. For example, the equation

$$x^2 + 1 = 0$$

has no solution in $\mathbb{R}$, since no real number has square -1. To overcome this limitation, one introduces a new number i satisfying

$$i^2 = -1,$$

and defines the *complex numbers* as all expressions of the form

$$z = x + iy, \qquad x, y \in \mathbb{R}.$$

The set of complex numbers is denoted by $\mathbb{C}$. Every complex number has a *real part* $\Re(z) = x$ and an *imaginary part* $\Im(z) = y$.

Modulus and argument

For a complex number $z = x + iy$, its *modulus* is

$$|z| = \sqrt{x^2 + y^2},$$

and its *argument* is the angle θ (with $\theta \in \mathbb{R}$) such that

$$\cos\theta = \frac{x}{|z|}, \qquad \sin\theta = \frac{y}{|z|}.$$

The modulus represents the distance of z from the origin in the complex plane, and the argument represents the direction of z.

Trigonometric form

Every nonzero complex number $z = x + iy$ can be written in *trigonometric form*:

$$z = |z|\big(\cos\theta + i\sin\theta\big),$$

where $|z|$ is the modulus and θ is the argument of z. This representation is fundamental for multiplication, division, powers, and roots of complex numbers.

Examples

- $z = 3 + 3i$ has $|z| = 3\sqrt{2}$ and $\theta = \pi/4$, so
$$z = 3\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right).$$
- $z = -2i$ has $|z| = 2$ and $\theta = -\frac{\pi}{2}$.

De Moivre's formula

For every complex number written in trigonometric form
$$z = |z|(\cos\theta + i\sin\theta),$$
and for every integer $n \geq 1$, the powers of z are given by *De Moivre's formula*:
$$z^n = |z|^n(\cos(n\theta) + i\sin(n\theta)).$$
This identity shows that raising a complex number to a power multiplies its argument by n and raises its modulus to the n-th power.

Roots

De Moivre's formula also provides the n-th roots of a nonzero complex number. If
$$z = |z|(\cos\theta + i\sin\theta),$$
then the n distinct n-th roots of z are
$$\sqrt[n]{z}_k = |z|^{1/n}(\cos\frac{\theta + 2k\pi}{n} + i\sin\frac{\theta + 2k\pi}{n}), \qquad k = 0, 1, \ldots, n-1.$$
Each root corresponds to dividing the argument by n and adding the full rotations $\frac{2k\pi}{n}$.

Examples

- For $z = \cos\theta + i\sin\theta$,
$$z^5 = \cos(5\theta) + i\sin(5\theta).$$
- The cube roots of 1 are
$$\cos\frac{2k\pi}{3} + i\sin\frac{2k\pi}{3}, \qquad k = 0, 1, 2.$$

n-th roots of a complex number

Let z be a nonzero complex number written in trigonometric form
$$z = |z|(\cos\theta + i\sin\theta).$$
We want to determine all complex numbers w such that $w^n = z$, where $n \geq 1$ is an integer. A complex number w is an n-th root of z if and only if it has modulus
$$|w| = |z|^{1/n}$$
and argument
$$\arg(w) = \frac{\theta + 2k\pi}{n}, \qquad k = 0, 1, \ldots, n-1.$$
Thus the n distinct n-th roots of z are
$$w_k = |z|^{1/n}\left(\cos\frac{\theta + 2k\pi}{n} + i\sin\frac{\theta + 2k\pi}{n}\right), \qquad k = 0, 1, \ldots, n-1.$$
These roots are equally spaced on the circle of radius $|z|^{1/n}$ in the complex plane, forming a regular n-gon centered at the origin.

Example

The fourth roots of 1 are

$$\cos\frac{2k\pi}{4} + i\sin\frac{2k\pi}{4}, \qquad k = 0, 1, 2, 3,$$

that is,

$$1, \quad i, \quad -1, \quad -i.$$

9 Fundamental theorem of algebra

Complex numbers allow us to solve all polynomial equations. In contrast, the real numbers are not sufficient: for example, the equation

$$x^2 + 1 = 0$$

has no real solution. The introduction of the complex number i with $i^2 = -1$ makes it possible to solve every polynomial equation of positive degree.

Theorem 2 (Fundamental theorem of algebra)**.** *Every nonconstant polynomial with complex coefficients has at least one complex root.*

Equivalently, if

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0, \qquad a_n \neq 0,$$

is a polynomial of degree $n \geq 1$, then there exists $z_0 \in \mathbb{C}$ such that $p(z_0) = 0$.

Consequences

The theorem implies several fundamental facts:

- Every polynomial of degree n has exactly n complex roots, counted with multiplicity.
- Every polynomial can be factored completely over $\mathbb{C}$:

$$p(z) = a_n(z - z_1)(z - z_2)\cdots(z - z_n).$$

- The field $\mathbb{C}$ is *algebraically closed*: all algebraic equations can be solved within $\mathbb{C}$.

Examples

- The polynomial $z^2 + 1$ has the roots i and $-i$.
- The polynomial $z^3 - 1$ has the three roots

$$1, \qquad \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}, \qquad \cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}.$$

10 Functions of one variable

A *function* is a fundamental object in calculus. It assigns to each real number in a given set exactly one real number. Functions describe how quantities depend on one another and are represented geometrically in the Cartesian plane.

Definition of a function

Definition 13 (Function)**.** Let $A, B \subseteq \mathbb{R}$. A *function* is a rule

$$f : A \to B$$

that assigns to every $x \in A$ a unique real number $f(x) \in B$. The set A is the *domain* of f, the set B is the *codomain*, and the value $f(x)$ is the *image* of x.

Thus a function is a special kind of relation: every input has exactly one output. The collection of all pairs $(x, f(x))$ is the *graph* of f:

$$\operatorname{graph}(f) = \{(x, f(x)) : x \in A\} \subseteq \mathbb{R}^2.$$

The Cartesian plane

The *Cartesian plane*, introduced by René Descartes, is the set of all ordered pairs of real numbers:

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}.$$

It is built from two perpendicular axes:

- the horizontal axis (the x-axis), representing the input variable;
- the vertical axis (the y-axis), representing the output variable.

The axes intersect at the *origin* $(0, 0)$. Every point (x, y) in the plane corresponds to a unique pair of real numbers.

Graph of a function

The graph of a function f is the set of points $(x, f(x))$ in the Cartesian plane. It shows how the output varies as the input changes. A curve in the plane represents a function if and only if no vertical line intersects it more than once. This is the *vertical line test*, expressing that each x has exactly one $f(x)$.

Examples

- $f(x) = x^2$: its graph is a parabola opening upward.
- $f(x) = \frac{1}{x}$: its graph has two branches, one in each of the first and third quadrants.
- $f(x) = \sin x$: its graph is a periodic wave oscillating between -1 and 1.

10.1 Trigonometric functions

The trigonometric functions sin, cos, and tan are defined using the geometry of the unit circle in the Cartesian plane. For an angle θ, measured from the positive x-axis, the point on the unit circle has coordinates

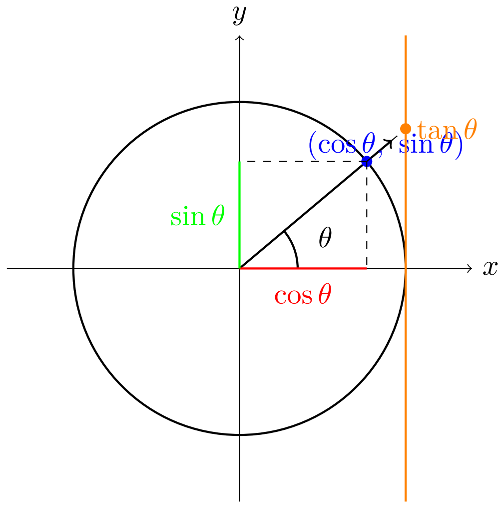
$$(\cos\theta,\ \sin\theta).$$

The tangent function is defined as

$$\tan\theta = \frac{\sin\theta}{\cos\theta},$$

whenever $\cos\theta \neq 0$.

Geometric representation



Interpretation

- $\cos\theta$ is the horizontal coordinate of the point on the unit circle.
- $\sin\theta$ is the vertical coordinate.
- $\tan\theta$ is the length of the segment where the ray at angle θ intersects the vertical line $x = 1$.

Thus the trigonometric functions arise naturally from the geometry of the unit circle.

10.2 Inverse functions

A function may admit an inverse only when it is *injective*: different inputs must produce different outputs. Formally, a function $f : A \to B$ is injective when

$$f(x_1) = f(x_2) \;\Rightarrow\; x_1 = x_2.$$

Injectivity ensures that every value in the image of f corresponds to exactly one input. In this case one can define the *inverse function* f^{-1}, which reverses the action of f.

Definition 14 (Inverse function)**.** Let $f : A \to B$ be an injective function. The *inverse function* of f is the map

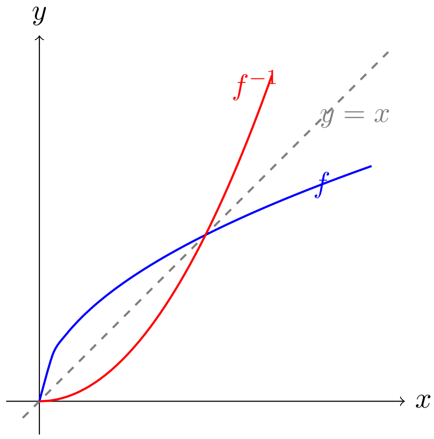
$$f^{-1} : f(A) \to A$$

satisfying

$$f^{-1}(f(x)) = x \qquad \text{for all } x \in A.$$

The graph of f^{-1} is obtained by reflecting the graph of f across the line $y = x$, called the *bisector* of the first and third quadrants.

Geometric representation



The bisector $y = x$ shows that the inverse function is obtained by exchanging the roles of input and output: the graph of f^{-1} is the reflection of the graph of f across this line.

11 Limits and continuity

Many concepts in calculus are based on the idea of approaching a value without necessarily reaching it. The simplest objects that illustrate this behaviour are *sequences* of real numbers. They provide the first setting in which limits are defined and studied.

11.1 Sequences

A *sequence* is an ordered list of real numbers

$$(a_n)_{n\geq 0} = a_0, a_1, a_2, \ldots$$

indexed by the integers $n \geq 0$. Formally, a sequence is a function

$$a : \mathbb{N} \to \mathbb{R}, \qquad n \mapsto a_n,$$

where $\mathbb{N}$ denotes the set of non'negative integers. The value a_n is called the *n'th term* of the sequence. A sequence may be described by an explicit formula (for example $a_n = 1/n$) or by a recurrence relation (for example $a_{n+1} = q\, a_n$).

11.2 Convergence of a sequence

A sequence may approach a real number as n becomes large. This behaviour is captured by the notion of *limit*.

Definition 15 (Limit of a sequence)**.** A sequence (a_n) *converges* to a real number L if for every $\varepsilon > 0$ there exists an integer N such that

$$|a_n - L| < \varepsilon \qquad \text{for all } n \geq N.$$

In this case one writes

$$\lim_{n\to\infty} a_n = L.$$

If no such real number L exists, the sequence is said to *diverge*.

11.3 Eventual behaviour of a sequence

Many properties of sequences do not need to hold for every index. It is often enough that they hold *eventually*, that is, from some point onward.

Definition 16 (Eventually)**.** Let (a_n) be a sequence of real numbers. A property $P(n)$ is said to hold *eventually* (or *for all sufficiently large* n) if there exists an integer N such that

$$P(n) \qquad \text{holds for all } n \geq N.$$

In other words, a sequence may fail to satisfy $P(n)$ for the first few terms, but as long as the property holds for all indices beyond some threshold N, it is considered to hold eventually. This notion is fundamental in limit theory, where statements about a_n approaching a value concern the behaviour of the sequence for large n, not its initial terms.

11.4 Types of behaviour of a sequence

A sequence may exhibit different kinds of behaviour as the index n becomes large. The standard distinction is the following.

Convergent sequences

A sequence (a_n) is *convergent* if it has a real limit, that is, if there exists $L \in \mathbb{R}$ such that

$$\lim_{n\to\infty} a_n = L.$$

Sequences divergent to infinity

A sequence is said to *diverge to* $+\infty$ if its terms grow without bound, meaning that for every $M \in \mathbb{R}$ there exists N such that

$$a_n > M \qquad \text{for all } n \geq N.$$

Similarly, it *diverges to* $-\infty$ if for every $M \in \mathbb{R}$ there exists N such that

$$a_n < M \qquad \text{for all } n \geq N.$$

In this sense, a "divergent sequence" is one whose terms tend to infinity in absolute value.

Sequences that are neither convergent nor divergent

A sequence may fail to converge and also fail to diverge to $\pm\infty$. Such sequences are described as *neither convergent nor divergent.* They may display oscillatory behaviour, irregular variation, or possess multiple accumulation points. These sequences do not approach any single real value, nor do they grow without bound.

11.5 Squeeze Theorem

The Squeeze Theorem provides a useful criterion for determining the limit of a sequence by comparing it with two other sequences whose limits are known.

Theorem 3 (Squeeze Theorem)**.** *Let (a_n), (b_n), and (c_n) be sequences of real numbers such that*

$$a_n \leq b_n \leq c_n \qquad \textit{for all sufficiently large } n.$$

If

$$\lim_{n\to\infty} a_n = L \qquad \textit{and} \qquad \lim_{n\to\infty} c_n = L,$$

then the sequence (b_n) also converges to L, that is,

$$\lim_{n\to\infty} b_n = L.$$

The theorem states that if a sequence is eventually trapped between two convergent sequences sharing the same limit, then it must converge to that limit as well.

11.6 The exponential limit and the number e

The number e plays a central role in calculus and analysis. It is defined through the behaviour of a specific sequence.

Theorem 4 (Limit defining the number e)**.** *The sequence*

$$a_n = \left(1 + \frac{1}{n}\right)^n$$

is increasing and bounded above. Therefore it is convergent, and its limit is denoted by e:

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e.$$

The number e is thus characterized as the unique real value approached by the sequence $\left(1 + \frac{1}{n}\right)^n$ as n becomes large. This limit provides the fundamental definition of the base of the natural logarithm.

11.7 Asymptotic comparisons and estimates

Asymptotic analysis studies the behaviour of sequences for large values of the index. It provides tools to compare the growth of different sequences and to obtain useful approximations.

Asymptotic comparison

Let (a_n) and (b_n) be sequences of real numbers. We say that a_n and b_n have the same asymptotic behaviour if

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 1.$$

In this case we write

$$a_n \sim b_n,$$

and say that a_n is *asymptotically equivalent* to b_n. This relation expresses that the two sequences grow at the same rate, even if their initial terms may differ.

Asymptotic estimates

A sequence (a_n) is said to be *of smaller order* than (b_n) if

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 0,$$

and we write

$$a_n = o(b_n).$$

This means that a_n grows strictly slower than b_n. Similarly, we say that a_n is *bounded in order* by b_n if there exists a constant $C > 0$ such that

$$|a_n| \leq C\,|b_n| \qquad \text{for all sufficiently large } n.$$

In this case we write

$$a_n = O(b_n).$$

These notations allow us to express growth rates and approximations in a concise and effective way.

Asymptotic inequalities

Asymptotic comparisons often rely on inequalities that hold eventually. If for sufficiently large n we have

$$a_n \leq b_n,$$

then the growth of (a_n) is asymptotically controlled by that of (b_n). Such inequalities are frequently used to establish limits or to derive asymptotic estimates.

11.8 Ratio Test

The ratio test provides a useful criterion for determining whether a sequence diverges to $+\infty$ or converges to 0, based on the behaviour of the ratio between consecutive terms.

Theorem 5 (Ratio Test)**.** *Let (a_n) be a sequence of positive real numbers and suppose that the limit*

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$$

exists.

- *If $L < 1$, then the sequence (a_n) converges to 0.*
- *If $L > 1$, then the sequence (a_n) diverges to $+\infty$.*
- *If $L = 1$, the test is inconclusive: the sequence may converge, diverge, or display irregular behaviour.*

The ratio test compares the growth of consecutive terms. If each term is eventually smaller than the previous one by a fixed proportion, the sequence must tend to zero. If each term is eventually larger by a fixed proportion, the sequence grows without bound.

11.9 Uniqueness of the limit

The limit of a sequence, when it exists, is uniquely determined. This fundamental property ensures that a sequence cannot converge to two different real numbers.

Theorem 6 (Uniqueness of the limit)**.** *Let (a_n) be a sequence of real numbers. If (a_n) converges to L and also converges to M, then*

$$L = M.$$

In other words, a convergent sequence cannot have two distinct limits. The eventual behaviour of the sequence determines a single real value, which is its unique limit.

11.10 Right and left limits of a function

When studying the behaviour of a function near a point, it is often necessary to distinguish between approaching the point from the left or from the right.

Definition 17 (Right limit)**.** Let f be a real function and let x_0 be a point in its domain. The *right limit* of f at x_0 is the value L such that

$$\lim_{x\to x_0^+} f(x) = L,$$

meaning that $f(x)$ approaches L as x approaches x_0 from values greater than x_0.

Definition 18 (Left limit)**.** The *left limit* of f at x_0 is the value M such that

$$\lim_{x\to x_0^-} f(x) = M,$$

meaning that $f(x)$ approaches M as x approaches x_0 from values smaller than x_0.

Existence of the limit

A function has a limit at a point if and only if both one-sided limits exist and are equal.

Theorem 7 (Existence of the limit)**.** *Let f be a real function and let x_0 be a point in its domain. The limit*

$$\lim_{x \to x_0} f(x)$$

exists and equals L if and only if

$$\lim_{x \to x_0^-} f(x) = L \qquad \text{and} \qquad \lim_{x \to x_0^+} f(x) = L.$$

Thus, the behaviour of the function near x_0 is fully determined by its left-hand and right-hand limits. If these two values coincide, the function has a well-defined limit at x_0.

Jump discontinuity

A function may fail to be continuous at a point because its left-hand and right-hand limits exist but are different. This type of discontinuity is called a *jump discontinuity*.

Definition 19 (Jump discontinuity)**.** Let f be a real function and let x_0 be a point in its domain. We say that f has a *jump discontinuity* at x_0 if the one-sided limits

$$\lim_{x \to x_0^-} f(x) \qquad \text{and} \qquad \lim_{x \to x_0^+} f(x)$$

both exist as finite real numbers, but

$$\lim_{x \to x_0^-} f(x) \neq \lim_{x \to x_0^+} f(x).$$

In this case the function "jumps" from one value to another when crossing the point x_0. The function cannot be continuous at x_0, since continuity requires the left-hand and right-hand limits to coincide and equal the value of the function.

11.11 Neighbourhood of a point

The behaviour of a function near a point is often described using the notion of a neighbourhood. A neighbourhood specifies a small interval around a point, without necessarily including the point itself.

Definition 20 (Neighbourhood)**.** Let $x_0 \in \mathbb{R}$ and let $\delta > 0$. The *neighbourhood* of x_0 of radius δ is the interval

$$(x_0 - \delta,\ x_0 + \delta).$$

It contains all real numbers whose distance from x_0 is less than δ.

Definition 21 (Punctured neighbourhood)**.** The *punctured neighbourhood* of x_0 of radius δ is the set

$$(x_0 - \delta,\ x_0 + \delta) \setminus \{x_0\}.$$

It includes all points sufficiently close to x_0, but excludes x_0 itself.

Neighbourhoods and punctured neighbourhoods are used to describe limits and continuity, since limits concern the behaviour of a function near a point rather than at the point itself.

11.12 Change of variable in limits

When evaluating limits, it is often useful to replace the variable with another expression that approaches the same value. The following theorem formalizes this idea.

Theorem 8 (Change of variable in limits)**.** *Let f be a real function and let g be another function such that*

$$\lim_{x \to x_0} g(x) = L.$$

If the limit

$$\lim_{y \to L} f(y)$$

exists and equals M, then the composite limit

$$\lim_{x \to x_0} f(g(x))$$

also exists and equals M.

In other words, if $g(x)$ approaches L as x approaches x_0, and $f(y)$ approaches M as y approaches L, then the composition $f(g(x))$ approaches M as x approaches x_0. The change of variable allows us to evaluate limits by substituting the inner expression with its limiting value.

11.13 Continuity of elementary functions

Elementary functions encountered in calculus possess strong continuity properties. The following results summarize the fundamental facts.

Definition 22 (Continuity of elementary functions)**.** The basic elementary functions — polynomials, rational functions (where defined), exponential and logarithmic functions, trigonometric and inverse trigonometric functions — are continuous on every point of their domain.

Definition 23 (Continuity preserved by algebraic operations)**.** Let f and g be continuous at a point x_0. Then the functions

$$f + g, \qquad f - g, \qquad f \cdot g, \qquad \frac{f}{g} \text{ (if } g(x_0) \neq 0)$$

are also continuous at x_0.

Definition 24 (Continuity preserved by composition)**.** If f is continuous at x_0 and g is continuous at $f(x_0)$, then the composition $g \circ f$ is continuous at x_0:

$$\lim_{x \to x_0} g(f(x)) = g\big(\lim_{x \to x_0} f(x)\big).$$

These properties ensure that any function built from elementary functions using algebraic operations and composition remains continuous on its domain.

11.14 Fundamental limits

Certain limits play a central role in calculus and appear frequently in the study of derivatives, continuity, and asymptotic behaviour. These are known as *fundamental limits*.

- **Exponential limit**
$$\lim_{x \to 0} (1 + x)^{1/x} = e.$$
- **Logarithmic limit**
$$\lim_{x \to 0} \frac{\ln(1 + x)}{x} = 1.$$

- **Trigonometric limits**

$$\lim_{x \to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

- **Exponential growth limit**

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

- **Power limit**

$$\lim_{x \to 0} \frac{x^n}{x} = 0 \qquad \text{for every } n > 1.$$

These limits are used to compute derivatives of elementary functions and to establish many foundational results in analysis. They also serve as reference points for approximations and asymptotic estimates.

11.15 Theorem of the zeros

A continuous function on an interval cannot change sign without crossing the value zero. This fundamental result is known as the *Theorem of the zeros*.

Theorem 9 (Theorem of the zeros)**.** *Let f be a real function that is continuous on the closed interval $[a, b]$. If*

$$f(a) \cdot f(b) < 0,$$

that is, if $f(a)$ and $f(b)$ have opposite signs, then there exists at least one point $c \in (a, b)$ such that

$$f(c) = 0.$$

In other words, a continuous function that takes positive values at one endpoint of an interval and negative values at the other must vanish somewhere inside the interval. The function cannot "jump over" the value zero without violating continuity.

11.16 Weierstrass Theorem

A continuous function defined on a closed and bounded interval always reaches its largest and smallest values. This fundamental result is known as the *Weierstrass Theorem*.

Theorem 10 (Weierstrass Theorem)**.** *Let f be a real function that is continuous on the closed interval $[a, b]$. Then f is bounded on $[a, b]$, and there exist points*

$$x_{\text{min}},\ x_{\text{max}} \in [a, b]$$

such that

$$f(x_{\text{min}}) = \min_{x \in [a,b]} f(x), \qquad f(x_{\text{max}}) = \max_{x \in [a,b]} f(x).$$

In other words, a continuous function on a closed interval cannot "escape" to infinity and cannot approach its extrema without attaining them. The compactness of the interval ensures both boundedness and the existence of maximum and minimum values.

11.17 Intermediate Value Theorem

A continuous function on an interval cannot “skip” values. If it takes two different values at the endpoints of an interval, then it must take every value in between. This fundamental result is known as the *Intermediate Value Theorem.*

Theorem 11 (Intermediate Value Theorem)**.** *Let f be a real function that is continuous on the closed interval $[a, b]$. Let L be any real number between $f(a)$ and $f(b)$, that is,*

$$f(a) < L < f(b) \quad \text{or} \quad f(b) < L < f(a).$$

Then there exists at least one point $c \in (a, b)$ such that

$$f(c) = L.$$

In other words, a continuous function on an interval takes all intermediate values between its values at the endpoints. The function cannot jump over any value without violating continuity.

11.18 Existence and uniqueness of the nth root

The nth root of a positive real number is well defined and unique. This fundamental property ensures that the equation $x^n = y$ has exactly one positive solution.

Theorem 12 (Existence and uniqueness of the nth root)**.** *For every real number $y > 0$ and every integer $n \geq 1$, there exists one and only one real number $x > 0$ such that*

$$x^n = y.$$

This number is called the nth root of y and is denoted by $\sqrt[n]{y}$.

In other words, the function $x \mapsto x^n$ is continuous and strictly increasing on $(0, +\infty)$, and therefore it admits a unique inverse on this interval. This inverse function assigns to each positive y its unique positive nth root.

11.19 Monotonicity theorem for functions

A monotone function on an interval has well’defined one’sided limits at every point of the interval. Inside the interval these limits are finite, while at the endpoints they may be finite or infinite.

Theorem 13 (Monotonicity theorem)**.** *Let f be a real function that is monotone (increasing or decreasing) on the interval $[a, b]$.*

- *For every point $x_0 \in (a, b)$, both one’sided limits*

  $$\lim_{x \to x_0^-} f(x) \qquad \text{and} \qquad \lim_{x \to x_0^+} f(x)$$

  exist and are finite.

- *At the left endpoint a, the right’hand limit*

  $$\lim_{x \to a^+} f(x)$$

  exists and may be finite or $+\infty$ or $-\infty$.

- *At the right endpoint b, the left’hand limit*

  $$\lim_{x \to b^-} f(x)$$

  exists and may be finite or $+\infty$ or $-\infty$.

In other words, monotonicity ensures that the function has a well’defined behaviour near every point of the interval: inside the interval the function cannot oscillate, and at the endpoints it may diverge but never oscillate.

12 Differential Calculus

Differential calculus studies how functions change and how their behaviour can be described locally. Its central concept is the derivative, which measures the instantaneous rate of variation of a function with respect to its variable. Through limits, we develop tools to understand linear approximations, tangent lines, and the sensitivity of functions to small changes in input. This section introduces the derivative, explores its fundamental properties, and presents the main rules of differentiation. We also examine differentiability, the relationship between continuity and derivatives, and the applications of derivatives to monotonicity, extrema, and the qualitative analysis of functions.

12.1 Derivative of a function

Let f be a real function defined on a subset of $\mathbb{R}$. As stated earlier, a function is a rule $f : A \to B$ that assigns to every $x \in A$ a unique real number $f(x) \in B$. The derivative describes how $f(x)$ changes when x varies.

Definition 25 (Derivative)**.** The *derivative* of f at a point x_0 is the limit

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

if this limit exists.

The quotient

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

is the slope of the secant line through the points $(x_0, f(x_0))$ and $(x_0 + h, f(x_0 + h))$. The derivative $f'(x_0)$ is the slope of the tangent line obtained as the limit of these secant slopes.

12.2 Slope and tangent line

The tangent line to the graph of f at x_0 is the line

$$y = f(x_0) + f'(x_0)(x - x_0).$$

- If $f'(x_0) > 0$, the graph is increasing at x_0.
- If $f'(x_0) < 0$, the graph is decreasing at x_0.
- If $f'(x_0) = 0$, the tangent line is horizontal.

Angle of inclination

Every nonvertical line has an angle of inclination θ with the positive x-axis. The slope m and the angle θ are related by

$$m = \tan\theta.$$

Therefore, if the tangent line at x_0 makes an angle θ with the x-axis, then

$$f'(x_0) = \tan\theta.$$

A large absolute value of $f'(x_0)$ corresponds to a steep tangent line. If the tangent line is vertical, the angle is $\theta = \frac{\pi}{2}$ and the slope is undefined: the derivative does not exist at that point.

Elementary derivatives

The following table collects the basic derivatives of the standard elementary functions. Each formula holds for every x in the domain where the expression is defined.

Function $f(x)$	**Derivative** $f'(x)$
c (constant)	0
x	1
x^n (for $n \in \mathbb{Z}$)	nx^{n-1}
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
$\frac{1}{x}$	$-\frac{1}{x^2}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
e^x	e^x
a^x $(a > 0)$	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\log_a x$ $(a > 0)$	$\frac{1}{x \ln a}$

12.3 Corner points, cusps, and inflection points

The derivative describes the local behaviour of the graph of a function. Certain characteristic points occur when the derivative fails to exist or when the curvature of the graph changes.

Corner point

A point x_0 is a *corner point* of the graph of f if the one-sided derivatives

$$f'_-(x_0) = \lim_{h \to 0^-} \frac{f(x_0 + h) - f(x_0)}{h}, \qquad f'_+(x_0) = \lim_{h \to 0^+} \frac{f(x_0 + h) - f(x_0)}{h}$$

both exist but are different:

$$f'_-(x_0) \neq f'_+(x_0).$$

The tangent line has a sudden change of slope, producing a sharp angle on the graph.

Cusp

A point x_0 is a *cusp* if the one-sided derivatives diverge with opposite signs:

$$f'_-(x_0) = -\infty, \qquad f'_+(x_0) = +\infty,$$

or vice versa. The graph approaches a vertical tangent from both sides, but with incompatible directions, creating a pointed tip.

Inflection point

A point x_0 is an *inflection point* of f if the function is differentiable near x_0 and the concavity changes sign. Equivalently, the second derivative (when it exists) satisfies

$$f''(x) < 0 \text{ for } x < x_0, \qquad f''(x) > 0 \text{ for } x > x_0,$$

or the opposite. At an inflection point the tangent line exists and crosses the graph, marking the transition between concave and convex behaviour.

Summary

- Corner point: one-sided derivatives exist but are different.
- Cusp: one-sided derivatives diverge with opposite signs.
- Inflection point: concavity changes sign; the tangent line exists.

Continuity and differentiability

Differentiability is a stronger property than continuity. A function may be continuous without being differentiable, but every differentiable function is necessarily continuous.

Theorem 14 (Differentiability implies continuity)**.** *Let f be a function defined on an interval and let x_0 be a point in its domain. If f is differentiable at x_0, then f is continuous at x_0.*

Proof. If f is differentiable at x_0, the limit

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists and is finite. Multiplying both sides by h gives

$$f(x_0 + h) - f(x_0) = h\, f'(x_0) + o(h),$$

where $o(h) \to 0$ as $h \to 0$. Therefore,

$$\lim_{h \to 0} \big(f(x_0 + h) - f(x_0)\big) = 0,$$

which is exactly the definition of continuity at x_0. □

Continuity without differentiability

The converse is false: continuity does not imply differentiability. A function may be continuous at a point but fail to have a derivative there. Typical examples include:

- corner points (e.g. $f(x) = |x|$ at $x = 0$),
- cusps (e.g. $f(x) = x^{2/3}$ at $x = 0$),
- vertical tangents (e.g. $f(x) = \sqrt[3]{x}$ at $x = 0$).

Summary

- Differentiability $\Rightarrow$ continuity.
- Continuity $\not\Rightarrow$ differentiability.
- Non-differentiability may occur at corners, cusps, or vertical tangents.

12.4 Derivative of an inverse function

Let f be a real function defined on an interval. When f is strictly monotonic, it is invertible, and its inverse function f^{-1} is also defined on an interval. The derivative of the inverse is determined by the derivative of f.

Theorem 15 (Derivative of the inverse function)**.** *Let f be a continuous and strictly monotonic function on an interval, and assume that f is differentiable at a point x_0 with*

$$f'(x_0) \neq 0.$$

Then the inverse function f^{-1} is differentiable at $y_0 = f(x_0)$, and its derivative is

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

Equivalently, for every x in the domain of f,

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)}.$$

Proof. Since f is strictly monotonic, it admits an inverse f^{-1}. For y near $y_0 = f(x_0)$, write $y = f(x)$, so that $x = f^{-1}(y)$. Then

$$\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)}.$$

Taking the limit as $y \to y_0$ (equivalently $x \to x_0$) gives

$$(f^{-1})'(y_0) = \lim_{x \to x_0} \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{f'(x_0)}.$$

□

Interpretation

The derivative of the inverse function is the reciprocal of the derivative of the original function. Geometrically, the graph of f^{-1} is the reflection of the graph of f across the line $y = x$, and the slope transforms by inversion:

$$m_{f^{-1}} = \frac{1}{m_f}.$$

Example

For $f(x) = e^x$, the inverse is $f^{-1}(y) = \ln y$. Since $f'(x) = e^x$, we obtain

$$(\ln y)' = \frac{1}{y},$$

in agreement with the theorem.

12.5 Fermat's theorem

Critical points of a function occur where the derivative is zero or does not exist. Fermat's theorem gives a necessary condition for a local extremum when the derivative exists.

Theorem 16 (Fermat's theorem)**.** *Let f be a real function defined on an interval, and let x_0 be an interior point of the interval. If f has a local maximum or a local minimum at x_0 and f is differentiable at x_0, then*

$$f'(x_0) = 0.$$

Proof. Assume f has a local maximum at x_0. Then there exists a neighbourhood of x_0 such that

$$f(x) \le f(x_0) \qquad \text{for all } x \text{ near } x_0.$$

For $h > 0$ small,

$$\frac{f(x_0 + h) - f(x_0)}{h} \le 0,$$

and for $h < 0$ small,

$$\frac{f(x_0 + h) - f(x_0)}{h} \ge 0.$$

Taking the limits as $h \to 0^+$ and $h \to 0^-$ shows that the right and left derivatives exist and are both equal to 0. Thus $f'(x_0) = 0$. The case of a local minimum is analogous. □

Consequences

- If $f'(x_0) \neq 0$, then f cannot have a local extremum at x_0.
- If $f'(x_0) = 0$, the point x_0 is a *critical point*, but it may be:
  - a local maximum,
  - a local minimum,
  - or neither (e.g. an inflection point).
- If f is not differentiable at x_0, a local extremum may still occur (e.g. $f(x) = |x|$ at $x = 0$).

Example

For $f(x) = x^3$, we have $f'(x) = 3x^2$, so $f'(0) = 0$. However, $x = 0$ is not a local extremum: the function is increasing on both sides. This illustrates that $f'(x_0) = 0$ is a necessary but not sufficient condition for an extremum.

12.6 Mean Value Theorem (Lagrange's theorem)

The Mean Value Theorem is one of the fundamental results of differential calculus. It states that for every sufficiently regular function, there exists a point where the instantaneous rate of change equals the average rate of change over an interval.

Theorem 17 (Mean Value Theorem)**.** *Let f be a real function that is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), with $a < b$. Then there exists at least one point $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define the auxiliary function

$$g(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right).$$

The function g is continuous on $[a, b]$ and differentiable on (a, b), and satisfies $g(a) = g(b) = 0$. By Rolle's theorem, there exists $c \in (a, b)$ such that $g'(c) = 0$. Computing the derivative,

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

so $g'(c) = 0$ implies

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

□

Interpretation

The quantity

$$\frac{f(b) - f(a)}{b - a}$$

is the slope of the secant line joining the points $(a, f(a))$ and $(b, f(b))$. The theorem guarantees that somewhere between a and b, the tangent line has exactly the same slope. Thus the instantaneous rate of change equals the average rate of change at some point.

Consequences

- If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$.
- If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on $[a, b]$.
- If $f'(x) < 0$ for all $x \in (a, b)$, then f is strictly decreasing on $[a, b]$.

Example

For $f(x) = x^2$ on $[1, 3]$, the average rate of change is

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$$

Since $f'(x) = 2x$, the equation $2c = 4$ gives $c = 2$, which lies in $(1, 3)$, as guaranteed by the theorem.

12.7 Monotonicity test

The derivative provides a precise criterion to determine whether a function is increasing or decreasing on an interval. This result is known as the *monotonicity test*.

Theorem 18 (Monotonicity test)**.** *Let f be a real function that is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b).*

- *If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on $[a, b]$.*
- *If $f'(x) < 0$ for all $x \in (a, b)$, then f is strictly decreasing on $[a, b]$.*
- *If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$.*

Proof. Assume $f'(x) > 0$ for all $x \in (a, b)$. For any $x_1, x_2 \in (a, b)$ with $x_1 < x_2$, the Mean Value Theorem gives a point $c \in (x_1, x_2)$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Since $f'(c) > 0$ and $x_2 - x_1 > 0$, we obtain $f(x_2) > f(x_1)$, proving that f is strictly increasing. The decreasing and constant cases follow analogously. □

Interpretation

The sign of the derivative determines the behaviour of the function:

- Positive derivative $\Rightarrow$ the function rises.
- Negative derivative $\Rightarrow$ the function falls.
- Zero derivative $\Rightarrow$ the function is flat (constant).

Example

For $f(x) = x^3$, we have $f'(x) = 3x^2 \geq 0$ for all x. The derivative is never negative, and equals 0 only at $x = 0$. Thus f is increasing on every interval, even though the slope is horizontal at the origin.

12.8 L'Hospital's rule

L'Hospital's rule is a fundamental tool for evaluating limits that produce indeterminate forms of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Theorem 19 (L'Hospital's rule)**.** *Let f and g be real functions differentiable on an open interval (a, b), and suppose that*

$$\lim_{x\to c} f(x) = 0, \qquad \lim_{x\to c} g(x) = 0,$$

or

$$\lim_{x\to c} f(x) = \pm\infty, \qquad \lim_{x\to c} g(x) = \pm\infty,$$

for some point $c \in (a, b)$ (or c equal to an endpoint). Assume also that $g'(x) \neq 0$ for all x near c, and that the limit

$$\lim_{x\to c} \frac{f'(x)}{g'(x)}$$

exists (finite or infinite). Then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}.$$

Interpretation

When the quotient $\frac{f(x)}{g(x)}$ produces an indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the limit can be computed by differentiating the numerator and denominator separately. The rule does not apply to other indeterminate forms such as $0 \cdot \infty$, $\infty - \infty$, 1^∞, 0^0, or ∞^0 without additional transformations.

Example

Consider the limit

$$\lim_{x\to 0} \frac{\sin x}{x}.$$

Both numerator and denominator tend to 0, so the form is $\frac{0}{0}$. Applying L'Hospital's rule,

$$\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{\cos x}{1} = 1.$$

Another example

For

$$\lim_{x\to +\infty} \frac{e^x}{x^3},$$

both numerator and denominator tend to $+\infty$. Applying L'Hospital's rule repeatedly,

$$\frac{e^x}{x^3} \longrightarrow \frac{e^x}{3x^2} \longrightarrow \frac{e^x}{6x} \longrightarrow \frac{e^x}{6},$$

so the limit is $+\infty$.

12.9 Limit of the derivative and one-sided differentiability

The behaviour of the derivative near an endpoint determines whether the one-sided derivative exists at that endpoint.

Theorem 20 (One-sided derivative from the limit of the derivative)**.** *Let f be a real function defined on the interval $[a, b)$, with b possibly $+\infty$. Assume that:*

- *f is continuous at a;*
- *f is differentiable on (a, b);*
- *the limit*

  $$\lim_{x \to a^+} f'(x) = m$$

  exists, finite or infinite.

Then the right-hand derivative of f at a exists and satisfies

$$f'_+(a) = m.$$

Left-hand version

If f is defined on $(a, b]$, continuous at b, differentiable on (a, b), and

$$\lim_{x \to b^-} f'(x) = m,$$

then the left-hand derivative at b exists and equals m:

$$f'_-(b) = m.$$

12.10 Second derivative

The second derivative describes how the rate of change of a function varies. While the first derivative measures the slope of the tangent line, the second derivative measures the curvature of the graph.

Definition 26 (Second derivative)**.** Let f be a differentiable function on an interval. If the derivative f' is differentiable, the *second derivative* of f is

$$f''(x) = (f'(x))'.$$

Geometric meaning

The second derivative indicates how the graph bends:

- If $f''(x) > 0$, the graph is *concave up*: the tangent slope is increasing, and the curve bends upward like a cup.
- If $f''(x) < 0$, the graph is *concave down*: the tangent slope is decreasing, and the curve bends downward like an arch.
- If $f''(x_0) = 0$, the point x_0 may be an *inflection point*, where the concavity changes sign.

Thus the second derivative provides information about the shape of the graph and helps identify inflection points and classify critical points.

12.11 Continuity of convex and concave functions

Convexity and concavity impose strong regularity conditions on a function. In particular, a convex or concave function cannot have interior discontinuities.

Theorem 21 (Continuity of convex and concave functions)**.** *Let f be a real function defined on an interval I (open, closed, or half-open). If f is convex or concave on I, then f is continuous at every interior point of I. Possible discontinuities may occur only at the endpoints of I.*

Interpretation

A convex or concave function cannot "jump" inside the interval: its graph must bend smoothly without breaks. However, if the interval has endpoints (for example $[a, b)$ or $(a, b]$), the function may fail to be continuous at a or b, but never in the interior. Convexity and concavity therefore guarantee:

- continuity on the interior of the interval;
- no oscillations or jumps inside the interval;
- possible discontinuities only at the boundary points.

12.12 Convexity and derivatives

Convexity can be characterized through the behaviour of the derivative. A convex function has a non'decreasing derivative; a concave function has a non'increasing derivative.

Theorem 22 (Convexity and derivatives)**.** *Let f be a real function defined on an interval I and differentiable on the interior of I.*

- *f is convex on I if and only if its derivative f' is non-decreasing on I.*
- *f is concave on I if and only if its derivative f' is non-increasing on I.*

If f is twice differentiable, these conditions are equivalent to:

- $f''(x) \geq 0$ *for all x in I (convexity),*
- $f''(x) \leq 0$ *for all x in I (concavity).*

Geometric meaning

- For a convex function, the slope of the tangent line increases: the graph bends upward.
- For a concave function, the slope of the tangent line decreases: the graph bends downward.
- The sign of the second derivative describes the curvature of the graph.

Convexity therefore corresponds to "accelerating slope", while concavity corresponds to "decelerating slope".

12.13 Convexity and tangent lines

Convexity has a clear geometric interpretation in terms of tangent lines. For a convex function, every tangent line lies below the graph; for a concave function, every tangent line lies above it.

Theorem 23 (Convexity and tangent lines)**.** *Let f be a real function defined on an interval I and differentiable on the interior of I.*

- *If f is convex on I, then for every $x_0 \in I$ and every $x \in I$,*

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0).$$

That is, the tangent line at x_0 lies below the graph of f.

- *If f is concave on I, then for every $x_0 \in I$ and every $x \in I$,*

$$f(x) \leq f(x_0) + f'(x_0)(x - x_0).$$

That is, the tangent line at x_0 lies above the graph of f.

Geometric meaning

For a convex function:

- the graph bends upward;
- the slope of the tangent line increases;
- every tangent line touches the graph from below.

For a concave function:

- the graph bends downward;
- the slope of the tangent line decreases;
- every tangent line touches the graph from above.

This property is fundamental in optimization and in the geometric interpretation of convexity.

12.14 Inflection points and the second derivative

An inflection point is a point where the graph of a function changes concavity. The second derivative provides a useful criterion for detecting such points.

Theorem 24 (Inflection point and second derivative)**.** *Let f be a real function defined on an interval and twice differentiable on the interior of the interval. If x_0 is an inflection point of f, then the concavity changes sign at x_0, and therefore*

$$f''(x_0) = 0$$

or the second derivative does not exist at x_0. Conversely, if $f''(x_0) = 0$ and the second derivative changes sign when passing through x_0, then x_0 is an inflection point of f.

Geometric meaning

At an inflection point:

- the graph switches from concave up to concave down, or vice versa;
- the tangent line exists and crosses the graph;
- the second derivative typically vanishes, but this condition alone is not sufficient — the change of concavity is essential.

Thus the second derivative helps identify inflection points, but the decisive criterion is the change in concavity.

12.15 Inflection point and tangent line

An inflection point is a point where the function changes concavity. A key geometric property is that the tangent line at such a point intersects (or is crossed by) the graph.

Theorem 25 (The tangent line at an inflection point)**.** *Let f be a real function defined on an interval and differentiable at a point x_0. If x_0 is an inflection point of f, then the tangent line at x_0*

$$T(x) = f(x_0) + f'(x_0)(x - x_0)$$

is crossed by the graph of f at x_0. Equivalently, the function lies on one side of the tangent line for $x < x_0$ and on the opposite side for $x > x_0$.

Geometric meaning

At an inflection point:

- the concavity changes sign;
- the tangent line touches the graph at x_0;
- the graph passes from one side of the tangent to the other.

Thus, the tangent line is not a supporting line (as in convex or concave functions), but a line that the curve actually crosses.

12.16 Little-o notation

The little-o notation describes a quantity that becomes negligible compared to another as the variable approaches a point. It is used to express that one function goes to zero faster than another.

Definition 27 (Little-o notation)**.** Let $g(x)$ be a nonzero function near a point x_0. We say that

$$f(x) = o\big(g(x)\big) \quad \text{as } x \to x_0$$

if

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0.$$

Meaning

The expression $f(x) = o(g(x))$ means that $f(x)$ becomes **small much faster** than $g(x)$ as x approaches x_0. Equivalently, $f(x)$ is negligible compared to $g(x)$ in that limit. Typical examples:

- $x = o(1)$ as $x \to 0$ (because $x/1 \to 0$),
- $x^2 = o(x)$ as $x \to 0$ (because $x^2/x = x \to 0$),
- $\sin x - x = o(x)$ as $x \to 0$.

Little-o notation is widely used in Taylor expansions and in expressing error terms.

Little-o notation and linear approximation

Little-o notation is used to describe error terms that become negligible compared to a given quantity. For a differentiable function, it provides a precise way to express the accuracy of the linear approximation.

Definition 28 (Little-o notation)**.** We write

$$f(x) = o(g(x)) \quad \text{as } x \to x_0$$

if

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0.$$

This means that $f(x)$ becomes negligible compared to $g(x)$ near x_0.

Usage in linear approximation

If f is differentiable at x_0, then for a small increment dx we have the expansion

$$f(x_0 + dx) = f(x_0) + f'(x_0)\,dx + o(dx) \quad \text{as } dx \to 0.$$

The term $o(dx)$ represents the error of the linear approximation: it is much smaller than dx itself.

Differential

The differential of f at x_0 is defined as the linear part of the increment:

$$df(x_0) = f'(x_0)\,dx.$$

Thus the change in the function can be written as

$$f(x_0 + dx) - f(x_0) = df(x_0) + o(dx),$$

meaning that the differential gives the best linear approximation, and the little-o term measures the deviation from linearity.

12.17 Taylor–Maclaurin formula with Peano remainder

The Taylor expansion expresses a function as a polynomial plus an error term that becomes negligible at small scales. The Peano remainder gives a very compact way to describe this error.

Theorem 26 (Taylor formula with Peano remainder)**.** *Let f be a real function that is n times differentiable at a point x_0. Then, as $x \to x_0$,*

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + o((x - x_0)^n)\,.$$

If $x_0 = 0$, this becomes the Maclaurin expansion:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + o(x^n) \quad \text{as } x \to 0.$$

Meaning

The term

$$o((x - x_0)^n)$$

means that the remainder is **negligible compared to $(x - x_0)^{n**}$ as x approaches x_0. Thus the polynomial of degree n gives the best local approximation of the function near x_0, and the Peano remainder quantifies how small the error is.

12.18 Taylor formula with Lagrange remainder

The Taylor expansion expresses a function as a polynomial plus a remainder term that depends on a derivative evaluated at an intermediate point. The Lagrange form of the remainder gives an explicit bound on the error.

Theorem 27 (Taylor formula with Lagrange remainder)**.** *Let f be a real function that is $(n+1)$ times differentiable on an open interval containing x_0 and x. Then there exists a point ξ between x_0 and x such that*

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + R_n(x),$$

where the remainder is

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}.$$

Maclaurin version

If $x_0 = 0$, the expansion becomes

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}x^{n+1},$$

with ξ between 0 and x.

Meaning

The Lagrange remainder shows that the error behaves like the next derivative evaluated at some intermediate point. It provides a practical estimate for how accurate the Taylor polynomial is near x_0.

13 Series

A series is the sum of the terms of an infinite sequence. While a sequence lists numbers, a series adds them together. The central question is whether these infinite sums converge to a finite value or diverge.

Definition

Let (a_n) be a real sequence. The *series* associated with (a_n) is the infinite sum

$$\sum_{n=0}^{\infty} a_n.$$

The partial sums are defined by

$$S_N = \sum_{n=0}^{N} a_n.$$

Convergence

The series

$$\sum_{n=0}^{\infty} a_n$$

is said to *converge* if the sequence of partial sums (S_N) converges to a finite limit:

$$\lim_{N\to\infty} S_N = S \in \mathbb{R}.$$

In this case, S is called the *sum* of the series. If the limit does not exist or is infinite, the series *diverges*.

Meaning

A convergent series represents an infinite process that stabilizes to a finite value. A divergent series grows without bound or oscillates. Series are fundamental in analysis, especially for defining functions through power expansions and studying approximations.

13.1 Geometric series

A *geometric series* is the series associated with a geometric progression.

Definition 29 (Geometric series)**.** Let $(a_n)_{n\geq 0}$ be a geometric progression of ratio q, so that

$$a_n = a_0\, q^n \qquad (n \geq 0).$$

The *geometric series* with first term a_0 and ratio q is the series

$$\sum_{n=0}^{\infty} a_0\, q^n = a_0 + a_0 q + a_0 q^2 + \cdots .$$

The constant a_0 is the *first term* and q is the *ratio*.

By the geometric sum formula the partial sums are

$$S_N = \sum_{n=0}^{N} a_0\, q^n = a_0\, \frac{1-q^{N+1}}{1-q} \qquad (q \neq 1).$$

When $|q| < 1$ the partial sums converge to a finite limit, so the series converges and

$$\sum_{n=0}^{\infty} a_0\, q^n = \frac{a_0}{1-q}.$$

For $|q| \geq 1$ (with $a_0 \neq 0$) the series diverges.

13.2 Necessary condition for convergence

Theorem 28 (Necessary condition for convergence)**.** *If the series*

$$\sum_{n=0}^{\infty} a_n$$

converges, then its general term tends to zero:

$$\lim_{n\to\infty} a_n = 0.$$

The condition is necessary but not sufficient: the series may diverge even when its general term tends to zero. This gives a first divergence test: if $\lim_{n\to\infty} a_n \neq 0$, or the limit does not exist, then the series diverges. A classical warning is the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n},$$

whose general term tends to zero, while the series itself diverges.

13.3 Remainder of a convergent series

Definition 30 (Remainder of a series)**.** Let

$$\sum_{n=0}^{\infty} a_n$$

be a convergent series with sum S and partial sums S_N. For every $N \geq 0$, the *remainder* (or *tail*) of the series is

$$R_N = S - S_N = \sum_{n=N+1}^{\infty} a_n.$$

The absolute value $|R_N|$ is the *error* made when the sum S of the whole series is approximated by the partial sum S_N.

Theorem 29 (The remainder of a convergent series tends to zero)**.** *If the series $\sum_{n=0}^{\infty} a_n$ converges, then its remainder R_N tends to zero:*

$$\lim_{N\to\infty} R_N = 0.$$

The remainder measures what is left of the sum after the first $N+1$ terms. The theorem states that, for a convergent series, this leftover becomes arbitrarily small.

Proof. Since the series converges, the partial sums converge to the sum:

$$\lim_{N\to\infty} S_N = S.$$

By the definition of the remainder, $R_N = S - S_N$. Passing to the limit,

$$\lim_{N\to\infty} R_N = \lim_{N\to\infty} (S - S_N) = S - S = 0.$$

Therefore $\lim_{N\to\infty} R_N = 0$. □

From the identity $S = S_N + R_N$, the sum of a convergent series can be written as a partial sum plus a remainder:

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{N} a_n + \sum_{n=N+1}^{\infty} a_n.$$

In particular, this relation is often used to estimate the error of numerical approximations of the sum of a series.

13.4 Generalized harmonic series

The *generalized harmonic series* (also called the *p-series*) is the series

$$\sum_{n=1}^{\infty} \frac{1}{n^p},$$

where $p \in \mathbb{R}$ is a fixed exponent. For $p = 1$ it reduces to the harmonic series.

Theorem 30 (Convergence of the generalized harmonic series)**.** *The generalized harmonic series*

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$. For $p \leq 1$ the series diverges.

The theorem sharpens the warning of the harmonic series: the general term $1/n^p$ tends to zero for every $p > 0$, but that alone is not enough. Convergence requires the terms to go to zero fast enough, which happens exactly when $p > 1$. When it converges, the sum is denoted by $\zeta(p)$, the value of Euler's zeta function at p:

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \zeta(p).$$

The case $p = 1$ is the harmonic series and diverges; the case $p < 1$ diverges as well, since then

$$\frac{1}{n^p} \geq \frac{1}{n},$$

so its partial sums are bounded below by those of the harmonic series. *Proof idea for $p > 1$.* Group the terms in blocks

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{1}{n^p} &= 1 + \left(\frac{1}{2^p} + \frac{1}{3^p}\right) + \left(\frac{1}{4^p} + \cdots + \frac{1}{7^p}\right) + \cdots \\ &\leq 1 + \frac{2}{2^p} + \frac{4}{4^p} + \cdots = \sum_{k=0}^{\infty} \left(2^{1-p}\right)^k.\end{aligned}$$

The last series is geometric with ratio $2^{1-p} < 1$, hence it converges. Therefore the generalized harmonic series converges for every $p > 1$.

13.5 Criteria for series with non-negative or non-positive terms

Note. The criteria in this subsection apply to series $\sum a_n$ whose terms are eventually *non-negative* ($a_n \geq 0$) or eventually *non-positive* ($a_n \leq 0$). This means that all the terms of the series, except possibly a finite number of them, have the same sign. Since a finite number of terms do not affect the convergence or divergence of a series, the criteria remain valid if finitely many terms have the opposite sign. If the terms are eventually non-positive, one reduces to the non-negative case by applying the criterion to $-a_n$.

13.5.1 Root test

The *root test* (also called *Cauchy's root test*) decides the behaviour of a series with eventually non-negative terms by looking at the asymptotic growth of its terms.

Definition 31 (Root test)**.** Let $\sum a_n$ be a series with eventually non-negative terms, and let

$$\ell = \limsup_{n\to\infty} \sqrt[n]{a_n}$$

be the upper limit of the sequence of the n-th roots of a_n.

Theorem 31 (Root test)**.** *With the notation above,*

1. *if $\ell < 1$, the series* converges*;*
2. *if $\ell > 1$, the series* diverges*;*
3. *if $\ell = 1$, the test is* inconclusive*: the series may converge or diverge.*

Proof idea. If $\ell < 1$, choose a number r with $\ell < r < 1$. For all sufficiently large n,

$$\sqrt[n]{a_n} \leq r, \qquad \text{hence} \qquad a_n \leq r^n.$$

The series $\sum r^n$ is geometric with ratio $r < 1$, hence it converges, and by comparison $\sum a_n$ converges. If $\ell > 1$, then $\sqrt[n]{a_n} \geq 1$ infinitely often, so $a_n \geq 1$ infinitely often and the general term does not tend to zero; the series diverges. The case $\ell = 1$ is inconclusive: for example, the harmonic series $\sum \frac{1}{n}$ diverges and the generalized harmonic series $\sum \frac{1}{n^2}$ converges, yet in both cases $\ell = 1$.

13.5.2 Ratio test

The *ratio test* (also called *d'Alembert's ratio test*) studies the behaviour of a series with eventually positive terms by looking at the ratio between consecutive terms.

Theorem 32 (Ratio test)**.** *Let $\sum a_n$ be a series with $a_n > 0$ eventually, and suppose that the following limit exists (finite or infinite):*

$$\ell = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}.$$

1. *if $\ell < 1$, the series* converges*;*
2. *if $\ell > 1$, the series* diverges*;*
3. *if $\ell = 1$, the test is* inconclusive*: the series may converge or diverge.*

The ratio test is particularly effective when the general term is a product, a factorial, or an exponential, since then the ratio simplifies to a constant. *Proof idea.* If $\ell < 1$, choose a number r with $\ell < r < 1$. For all sufficiently large n,

$$\frac{a_{n+1}}{a_n} \le r,$$

so that

$$a_n \le C\, r^n$$

for some constant $C > 0$. The series $\sum r^n$ is geometric with ratio $r < 1$, hence it converges, and by comparison $\sum a_n$ converges. If $\ell > 1$, then eventually $a_{n+1} > a_n$, so the general term does not tend to zero and the series diverges. The case $\ell = 1$ is inconclusive: for example, the harmonic series $\sum \frac{1}{n}$ diverges and the generalized harmonic series $\sum \frac{1}{n^2}$ converges, yet in both cases $\ell = 1$.

13.5.3 Comparison test

The *comparison test* determines the behaviour of a series with non-negative terms by comparing it with another series whose behaviour is known.

Theorem 33 (Direct comparison test)**.** *Let $\sum a_n$ and $\sum b_n$ be two series such that eventually*

$$0 \le a_n \le b_n.$$

1. *If $\sum b_n$ converges, then $\sum a_n$ converges.*
2. *If $\sum a_n$ diverges, then $\sum b_n$ diverges.*

The direct comparison test applies verbatim to the series considered in this subsection, whose terms are eventually non-negative. For a series with eventually positive or non-positive terms, the same test applied to $-a_n$ reduces to this non-negative case. *Proof idea.* For non-negative terms, the partial sums are increasing. If $\sum b_n$ converges, then $S_N = \sum_{n=0}^{N} a_n \le \sum_{n=0}^{N} b_n \le \sum_{n=0}^{\infty} b_n$, so (S_N) is increasing and bounded above, hence convergent. The second statement is the contrapositive of the first.

Theorem 34 (Limit comparison test)**.** *Let $\sum a_n$ and $\sum b_n$ be two series with positive terms, and suppose that the limit*

$$L = \lim_{n\to\infty} \frac{a_n}{b_n}$$

exists. If $0 < L < \infty$, then $\sum a_n$ and $\sum b_n$ either both converge or both diverge. If $L = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges; if $L = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

In particular, if $a_n \sim b_n$, then the two series have the same behaviour. *Proof idea.* If $0 < L < \infty$, for all sufficiently large n,

$$\frac{L}{2} \le \frac{a_n}{b_n} \le 2L,$$

hence

$$\frac{L}{2}\, b_n \le a_n \le 2L\, b_n.$$

Applying the direct comparison test in both directions gives the result. The cases $L = 0$ and $L = \infty$ follow again from the direct comparison test. The generalized harmonic series of §13.4 is often used as a reference: to apply the direct comparison test, choose $b_n = 1/n^p$ with $p > 1$ to prove convergence, or with $p \le 1$ to prove divergence.

13.6 Series with varying signs

The criteria of §13.5 apply to series whose terms have the same sign, except possibly a finite number of them. Here we consider the opposite situation: series whose terms change sign indefinitely, without any restriction. For such series the criteria of §13.5 cannot be applied directly, and the notion of *absolute convergence* becomes central.

Definition 32 (Absolute convergence)**.** The series

$$\sum_{n=0}^{\infty} a_n$$

is said to *converge absolutely* if the series of the absolute values converges:

$$\sum_{n=0}^{\infty} |a_n| \text{ converges.}$$

If $\sum a_n$ converges but $\sum |a_n|$ diverges, the series is said to *converge conditionally.*

Absolute convergence is a stronger property than convergence: it imposes a condition not only on the sum of the terms, but on the sum of their absolute values.

Theorem 35 (Absolute convergence implies convergence)**.** *If the series*

$$\sum_{n=0}^{\infty} a_n$$

converges absolutely, then it converges:

$$\sum_{n=0}^{\infty} |a_n| \textit{ converges} \quad \Longrightarrow \quad \sum_{n=0}^{\infty} a_n \textit{ converges.}$$

Moreover,

$$\left|\sum_{n=0}^{\infty} a_n\right| \le \sum_{n=0}^{\infty} |a_n|.$$

The usefulness for series with varying signs is immediate: instead of studying the sign behaviour of a_n, one studies the series of $|a_n|$, whose terms are non-negative and therefore fall under the criteria of §13.5.

Proof. For every n, the terms $a_n + |a_n|$ satisfy

$$0 \le a_n + |a_n| \le 2\,|a_n|.$$

Since $\sum |a_n|$ converges, by the direct comparison test the series $\sum(a_n + |a_n|)$ converges. Then

$$a_n = (a_n + |a_n|) - |a_n|,$$

so $\sum a_n$ is the difference of two convergent series, hence it converges. For the inequality, apply the triangle inequality to the partial sums and take the limit:

$$\left|\sum_{n=0}^{N} a_n\right| \leq \sum_{n=0}^{N} |a_n|,$$

and passing to the limit as $N \to \infty$ gives the stated bound. □

A classical example of conditional convergence is the alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n},$$

which converges (by Leibniz's criterion) but does not converge absolutely, since $\sum 1/n$ diverges.

13.6.1 Leibniz's criterion

Leibniz's criterion (also called the *alternating series test*) deals with alternating series, i.e. series of the form

$$\sum_{n=0}^{\infty} (-1)^n a_n,$$

where the terms a_n are eventually non-negative.

Theorem 36 (Leibniz's criterion)**.** *Let (a_n) be a sequence of real numbers eventually non-negative. If (a_n) is eventually decreasing and*

$$\lim_{n\to\infty} a_n = 0,$$

then the alternating series

$$\sum_{n=0}^{\infty} (-1)^n a_n$$

converges.

Moreover, if the alternating series converges to S and S_N is its N-th partial sum, then the error is bounded by the first omitted term:

$$|S - S_N| = |R_N| \leq a_{N+1}.$$

The remainder R_N has the same sign as the first omitted term $(-1)^{N+1}a_{N+1}$, so the partial sums S_N oscillate around the sum S while getting closer to it. *Proof idea.* Assume $a_n \geq 0$ and decreasing. The even partial sums S_{2k} form an increasing sequence bounded above by S_1, and the odd partial sums S_{2k+1} form a decreasing sequence bounded below by S_0. Both therefore converge. Since

$$S_{2k+1} - S_{2k} = -a_{2k+1} \to 0,$$

the two limits coincide, and the common limit is the sum of the alternating series. Leibniz's criterion justifies the convergence of the alternating harmonic series shown above, and gives the quantitative estimate $|R_N| \leq \frac{1}{N+1}$ for its approximation.

13.6.2 Dirichlet's criterion

Dirichlet's criterion (or *Dirichlet's test*) is a convergence test for series of the form

$$\sum_{n=0}^{\infty} a_n b_n,$$

where the sequence (a_n) is monotone and tends to zero, and the partial sums of (b_n) are bounded. It generalizes Leibniz's criterion, which corresponds to the choice $b_n = (-1)^n$.

Theorem 37 (Dirichlet's criterion)**.** *Let (a_n) be a real sequence eventually monotone and such that*

$$\lim_{n\to\infty} a_n = 0,$$

and let (b_n) be a sequence (real or complex) whose partial sums

$$B_N = \sum_{n=0}^{N} b_n$$

are bounded: there exists $M > 0$ with $|B_N| \leq M$ for every $N \geq 0$. Then the series

$$\sum_{n=0}^{\infty} a_n b_n$$

converges.

The hypothesis on (a_n) can be relaxed: it suffices that (a_n) be of bounded variation, that is, that the series $\sum |a_{n+1} - a_n|$ converges. *Proof idea.* Write the general term using the partial sums of (b_n):

$$a_n b_n = a_n(B_n - B_{n-1}).$$

Summing by parts (Abel's transformation),

$$\sum_{n=0}^{N} a_n b_n = a_N B_N + \sum_{n=0}^{N-1} (a_n - a_{n+1}) B_n.$$

If (a_n) is decreasing to 0, then $a_N B_N \to 0$ and the remaining series converges, because the terms $(a_n - a_{n+1})$ are non-negative and sum to a_0, while B_n is bounded. The theorem follows. *Example.* For every $\theta \notin 2\pi\mathbb{Z}$, the series

$$\sum_{n=1}^{\infty} \frac{\sin(n\theta)}{n}, \qquad \sum_{n=1}^{\infty} \frac{\cos(n\theta)}{n}$$

converge, although the terms oscillate and their absolute values form the divergent harmonic series. Here $a_n = 1/n$ is decreasing to 0, and $b_n = \sin(n\theta)$, or $b_n = \cos(n\theta)$, has bounded partial sums, since they are given by the Dirichlet kernel formula.

13.7 Taylor series. Complex exponential

The Taylor formula of §12.17 expresses f as a polynomial plus a remainder. When the remainder vanishes in the limit, the polynomials extend to an infinite series.

Definition 33 (Taylor series)**.** Let f be infinitely differentiable at x_0. The *Taylor series* of f at x_0 is the series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n.$$

If $x_0 = 0$, it is called the *Maclaurin series.*

The Taylor series is a power series: its convergence region is an interval centered at x_0, and its radius of convergence can be determined with the ratio test (§13.5.2). When the Taylor series converges to $f(x)$ on an interval, that is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n,$$

the function is said to be *analytic* in that interval. *Examples.* The Maclaurin series of the elementary functions converge on the whole real line:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \qquad \cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}.$$

The series of e^x is the starting point for extending the exponential to the complex plane.

13.7.1 Complex exponential

Definition 34 (Complex exponential)**.** For every complex number z, the *complex exponential* is defined by the series

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

For every $z \in \mathbb{C}$ the series converges absolutely: the series of the moduli is $\sum |z|^n/n!$, which converges by the ratio test applied to the positive terms $|z|^n/n!$. *Series of the elementary trigonometric functions.* The Maclaurin series of the elementary trigonometric functions are the following:

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \qquad (x \in \mathbb{R}),$$

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \qquad (x \in \mathbb{R}),$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots \qquad \left(|x| < \frac{\pi}{2}\right),$$

$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \frac{5x^7}{112} + \cdots \qquad (|x| < 1),$$

$$\arctan x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{2k+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \qquad (|x| \le 1).$$

The series of $\sin x$ and $\cos x$ converge on the whole real line; those of $\tan x$, $\arcsin x$ and $\arctan x$ only on their interval of convergence indicated above. *Euler's formulas.* The complex exponential and the trigonometric functions are related by *Euler's formulas*: for every real x,

$$e^{ix} = \cos x + i \sin x \qquad \text{and} \qquad e^{-ix} = \cos x - i \sin x.$$

Equivalently, solving these identities for $\cos x$ and $\sin x$,

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \qquad \sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

The first pair is the polar form of a unit complex number; the second expresses the trigonometric functions through the complex exponential. Both pairs also follow from the Maclaurin series above, since e^{ix} and e^{-ix} combine the even powers into $\cos x$ and the odd powers into $\sin x$. For a general complex number $z = x + iy$,

$$e^z = e^{x+iy} = e^x(\cos y + i \sin y).$$

In particular, for $y = \pi$ one obtains Euler's identity

$$e^{i\pi} + 1 = 0,$$

and the complex exponential is periodic:

$$e^{i(y+2\pi)} = e^{iy}.$$

13.7.2 Complex logarithm

The complex exponential is periodic, hence it is not one-to-one on $\mathbb{C}$. Its inverse is therefore a multivalued function.

Definition 35 (Complex logarithm)**.** For a non-zero complex number z, the *complex logarithm* is defined as the inverse of the complex exponential:

$$w = \log z \qquad \Longleftrightarrow \qquad e^w = z, \qquad z \neq 0.$$

Writing $z = \rho e^{i\theta}$ and $w = u + iv$,

$$e^{u+iv} = \rho e^{i\theta} \qquad \Longleftrightarrow \qquad e^u = \rho \quad \text{and} \quad v = \theta + 2\pi k, \quad k \in \mathbb{Z}.$$

Therefore

$$\log z = \log |z| + i(\arg z + 2\pi k), \qquad k \in \mathbb{Z},$$

where $\arg z$ is any argument of z.

The complex logarithm is multivalued: for each $z \neq 0$ there are infinitely many values of $\log z$, differing by multiples of $2\pi i$.

Definition 36 (Principal value of the logarithm)**.** The *principal value* of the logarithm is obtained by choosing the principal argument $\operatorname{Arg} z \in (-\pi, \pi]$:

$$\operatorname{Log} z = \log |z| + i \operatorname{Arg} z.$$

This is a single-valued function, defined on $\mathbb{C} \setminus \{0\}$, continuous on $\mathbb{C} \setminus (-\infty, 0]$.

Examples. Some principal values:

$$\operatorname{Log} 1 = 0, \qquad \operatorname{Log} i = i\frac{\pi}{2}, \qquad \operatorname{Log}(-1) = i\pi, \qquad \operatorname{Log}(1-i) = \frac{1}{2}\log 2 - i\frac{\pi}{4}.$$

On the principal branch the usual properties hold when the arguments are chosen consistently: the logarithm of a product is the sum of the logarithms,

$$\operatorname{Log}(z_1 z_2) = \operatorname{Log} z_1 + \operatorname{Log} z_2,$$

up to possibly adding a multiple of $2\pi i$.

14 Integral calculus for functions of one variable

Integral calculus studies the inverse operation of differentiation and the process of accumulating a quantity from its rate of variation. Its central concept is the *integral*, which appears in two related forms: the *indefinite integral*, the family of antiderivatives of a function, and the *definite integral*, the signed area between the graph of a function and the x-axis over an interval. While the derivative answers the question "how fast does a function change?", the integral answers the complementary question "how much has been accumulated, or what is the total size of the region under a curve?". The bridge between the two is the *Fundamental Theorem of Calculus*, which states that differentiation and integration are inverse operations of one another. In this sense, derivatives and integrals are the two sides of the same coin, and a large part of the section is devoted to making this relationship precise. This section introduces the concept of antiderivative, the Riemann definite integral and its main properties, and the fundamental theorems that connect integration with differentiation. We then examine the standard integration techniques, improper integrals, and the typical applications of the definite integral: areas between curves, volumes of solids of revolution, and lengths of curves.

14.1 Cauchy–Riemann sums

The definite integral is motivated by the problem of measuring the area of the region between the graph of a function and the horizontal axis. The idea is to approximate this region with rectangles, whose areas are easy to compute, and then to improve the approximation indefinitely. The construction goes back to Cauchy and Riemann, and the sums that appear in it are called *Cauchy–Riemann sums*.

Definition 37 (Partition)**.** Let $[a, b]$ be a closed interval with $a < b$. A *partition* of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_n = b.$$

The intervals $[x_{i-1}, x_i]$, for $i = 1, \ldots, n$, are the *subintervals* of the partition, and its *mesh* is the length of the longest subinterval:

$$\|P\| = \max_{1\leq i\leq n} (x_i - x_{i-1}).$$

In each subinterval a point $\xi_i \in [x_{i-1}, x_i]$ is chosen; such a point is called a *sampling point*. The value $f(\xi_i)$ is used as the height of the rectangle that approximates the area over $[x_{i-1}, x_i]$.

Definition 38 (Cauchy–Riemann sum)**.** Let f be a real function defined on $[a, b]$, let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$, and let $\xi_i \in [x_{i-1}, x_i]$ be a sampling point in each subinterval. The *Cauchy–Riemann sum* of f relative to P and to the sampling points ξ_i is

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)\,(x_i - x_{i-1}).$$

Each term $f(\xi_i)(x_i - x_{i-1})$ is the area of the rectangle with base $x_i - x_{i-1}$ and height $f(\xi_i)$. The sum is therefore the signed area of the union of these rectangles: it approximates the area between the graph of f and the axis over $[a, b]$, counted as negative where f is negative. The choice of the sampling points gives common families of sums: if $\xi_i = x_{i-1}$ one obtains the *left sums*, if $\xi_i = x_i$ the *right sums*, and if $\xi_i = (x_{i-1} + x_i)/2$ the *midpoint sums*. A frequently used case is the *uniform partition*, in which $[a, b]$ is divided into n subintervals of equal length. Setting

$$h = \frac{b-a}{n}, \qquad x_i = a + i\,h \quad (i = 0, 1, \ldots, n),$$

every subinterval has the same length $x_i - x_{i-1} = h = (b - a)/n$, and the Cauchy–Riemann sum takes the form

$$S_n = \sum_{i=1}^{n} f(\xi_i)\,\frac{b-a}{n} = \frac{b-a}{n}\sum_{i=1}^{n} f(\xi_i).$$

The mesh of such a partition is $\|P\| = h = (b - a)/n$, which tends to 0 exactly when $n \to \infty$; the definition of the integral then reads

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n}\sum_{i=1}^{n} f(\xi_i).$$

When the sampling points are the right endpoints $\xi_i = x_i$, this is the classical *Cauchy sum*, the form originally used to introduce the integral.

Definition 39 (Definite integral)**.** Let f be bounded on $[a, b]$. If, as the mesh of the partition tends to 0 and independently of the choice of the sampling points, the Cauchy–Riemann sums $S(f, P, \xi)$ all tend to the same finite limit I, then f is said to be *Riemann integrable* on $[a, b]$, and the limit is called the *definite integral* of f over $[a, b]$:

$$\int_a^b f(x)\,dx = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(\xi_i)\,(x_i - x_{i-1}).$$

Continuous functions on a closed interval are Riemann integrable; the same is true for the larger class of bounded functions whose set of discontinuities has negligible measure.

Meaning

The Cauchy–Riemann sum is the total signed area of a collection of rectangles that imitate the graph of f. Refining the partition makes the imitation more faithful, and the limit of the sums as the mesh goes to zero is the signed area of the region under the graph, that is, the definite integral.

14.2 Classes of integrable functions

Riemann integrability is a restrictive property, but three large and practically important classes of functions always satisfy it: continuous functions, monotone functions, and bounded functions with only finitely many points of discontinuity. In each case the function is defined on a closed and bounded interval $[a, b]$.

Theorem 38 (Continuous functions). *If f is continuous on $[a, b]$, then f is Riemann integrable on $[a, b]$.*

Theorem 39 (Monotone functions). *If f is monotone on $[a, b]$, then f is Riemann integrable on $[a, b]$.*

Theorem 40 (Bounded functions with finitely many discontinuities). *If f is bounded on $[a, b]$ and discontinuous at only finitely many points, then f is Riemann integrable on $[a, b]$.*

These conditions are sufficient but not necessary: the class of Riemann integrable functions is strictly larger, and it includes in particular every bounded function whose set of discontinuity points has negligible measure.

Meaning

These three classes guarantee that the definite integral is well defined for the functions that appear in practice. Continuous, monotone and piecewise-continuous functions all admit an integral, so in the rest of the section we can freely use the integral sign without worrying about whether the object it denotes exists.

14.3 Mean value theorem for integrals

The definite integral measures the accumulation of a quantity over an interval. By the average value, the integral can be read back as a kind of "mean height" of the function: the mean value theorem for integrals makes this interpretation precise under continuity.

Theorem 41 (Mean value theorem for integrals). *Let f be continuous on $[a, b]$. Then there exists a point $c \in [a, b]$ such that*

$$\int_a^b f(x)\,dx = f(c)\,(b-a).$$

Dividing by $b - a$, the point c realises the *average value* of f on $[a, b]$:

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

The area under the graph of f is therefore equal to the area of the rectangle with base $b - a$ and height $f(c)$. When a weighted average is needed, the following generalisation is used.

Theorem 42 (Generalised mean value theorem for integrals). *Let f be continuous on $[a, b]$ and let g be Riemann integrable on $[a, b]$ and of constant sign on $[a, b]$. Then there exists a point $c \in [a, b]$ such that*

$$\int_a^b f(x)\,g(x)\,dx = f(c)\int_a^b g(x)\,dx.$$

With $g \equiv 1$ the generalised statement reduces to the mean value theorem for integrals above.

Meaning

The mean value theorem for integrals guarantees that a continuous function attains its average value at some point of the interval: the integral can be replaced by the product of a single value $f(c)$ and the length $b - a$. This is the integral analogue of the mean value theorem of differential calculus, and it is useful every time an integral over an interval has to be converted back into a point value.

14.4 Fundamental theorem of calculus

The fundamental theorem of calculus connects the two operations that define calculus: differentiation and integration. It states, in two complementary parts, that integration and differentiation are inverse operations: the derivative of an integral returns the original function, and the integral of a derivative measures the total increment of a function. To state the first part, consider the *area function* of a continuous function f: the integral of f from a fixed point a to a variable point x,

$$F(x) = \int_a^x f(t)\,dt.$$

As x varies, $F(x)$ measures how much area has been accumulated between a and x.

Theorem 43 (Fundamental theorem of calculus, part I)**.** *Let f be continuous on $[a, b]$, and let*

$$F(x) = \int_a^x f(t)\,dt, \qquad x \in [a, b].$$

Then F is differentiable on $[a, b]$ and

$$F'(x) = f(x).$$

Thus the area function F is an antiderivative of f: differentiating the integral gives back the original function. In particular, every continuous function admits an antiderivative, even if no elementary formula for it is known. The first part reconstructs an antiderivative from an integral; the complementary second part, stated at the end of this section, reconstructs an integral from an antiderivative.

Meaning

The fundamental theorem of calculus is the reason why the two halves of calculus form a single theory. The derivative measures the rate of change of an accumulation process, and the integral measures the total change produced by a given rate. Part I guarantees that every continuous function possesses primitives and that the accumulation process can be differentiated back into the integrand.

14.5 Main methods of integration

By the fundamental theorem of calculus, computing a definite integral reduces to finding an antiderivative (also called a *primitive*) of the integrand. This subsection collects the main methods used for solving integrals: the table of immediate integrals, integration by parts,and substitution, together with linearity,which makes it possible to decompose integrals into simpler pieces.

14.5.1 Linearity

The indefinite integral is linear: for every pair of functions f, g and constants α, β,

$$\int (\alpha\, f(x) + \beta\, g(x))\,dx = \alpha \int f(x)\,dx + \beta \int g(x)\,dx.$$

Thus a primitive of a linear combination is the linear combination of the primitives, and integrals can be computed term by term.

14.5.2 Immediate integrals

The following table lists the primitives of the elementary functions. Each formula holds up to an arbitrary additive constant C; henceforth $\ln|x|$ is understood on every interval not containing 0.

$$
\begin{aligned}
&\int x^\alpha\,dx = \frac{x^{\alpha+1}}{\alpha+1} + C && (\alpha \neq -1),\\
&\int \frac{dx}{x} = \ln|x| + C, \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C && (a>0,\ a\neq 1),\\
&\int e^x\,dx = e^x + C, \qquad \int \sin x\,dx = -\cos x + C,\\
&\int \cos x\,dx = \sin x + C, \qquad \int \frac{dx}{\cos^2 x} = \tan x + C,\\
&\int \frac{dx}{\sin^2 x} = -\cot x + C, \qquad \int \frac{dx}{1+x^2} = \arctan x + C,\\
&\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C.
\end{aligned}
$$

14.5.3 Integration by parts

Integration by parts is the integral counterpart of the product rule of differentiation. If f and g are differentiable,and their derivatives are integrable, then

$$\int f(x)\,g'(x)\,dx = f(x)\,g(x) - \int f'(x)\,g(x)\,dx.$$

In differential notation,with $u = f(x)$,$v = g(x)$,this reads

$$\int u\,dv = u\,v - \int v\,du.$$

The method applies when the integrand is the product of a function that simplifies when differentiated, and a function whose primitive is known: choosing f so that f' is simpler than f usually improves the integral.

14.5.4 Substitution

Substitution is the integral counterpart of the chain rule of differentiation. If g has continuous derivative on the relevant interval,and F is a primitive of f, then

$$\int f(g(x))\,g'(x)\,dx = \int f(t)\,dt, \qquad t = g(x).$$

Equivalently,setting $t = g(x)$ and $dt = g'(x)\,dx$,the integral takes the following form after the change of variable. For definite integrals,the boundaries change accordingly:

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(t)\,dt,$$

when g has continuous derivative on $[a, b]$.

Meaning

These four tools cover the computation of most primitives of elementary functions:linearity decomposes the integral into simpler pieces, the table provides the atomic formulas,integration by parts handles products,and substitution handles compositions. In combination they reduce almost every integral that appears in exercises to the table of immediate integrals,after which the fundamental theorem of calculus gives the value of the corresponding definite integral.

14.6 Numerical calculation of an integral: the midpoint rule

A primitive of a function is not always available in explicit form, and even when it is, evaluating it can be cumbersome. In those cases the definite integral has to be computed *numerically*, that is, by approximating it with a finite number of evaluations of the integrand. The simplest numerical method is the *midpoint rule*, which is obtained from the Cauchy–Riemann sums of the subsection of the same name by choosing the midpoint of every subinterval as the sampling point.

Definition 40 (Midpoint rule)**.** Let f be a real function defined on $[a, b]$, and let $[a, b]$ be divided into n subintervals of equal length $h = (b - a)/n$, with division points

$$x_i = a + i\,h, \qquad i = 0, 1, \ldots, n.$$

The *midpoint rule* (or *rectangular rule*) approximates the definite integral of f over $[a, b]$ by

$$S_n = \frac{b-a}{n} \sum_{i=1}^{n} f\left(\frac{x_{i-1} + x_i}{2}\right),$$

where $(x_{i-1} + x_i)/2$ is the midpoint of the i-th subinterval $[x_{i-1}, x_i]$. Writing the midpoint as $x_{i-\frac{1}{2}} = (x_{i-1} + x_i)/2 = a + (i - \frac{1}{2})h$, the formula reads equivalently

$$S_n = h \sum_{i=1}^{n} f\left(x_{i-\frac{1}{2}}\right) = \frac{b-a}{n} \sum_{i=1}^{n} f\left(a + \left(i - \frac{1}{2}\right) \frac{b-a}{n}\right).$$

Geometrically, S_n is the sum of the areas of the n rectangles with base h and height equal to the value of f at the midpoint of each subinterval. The method is exact for affine functions, since the midpoint of an interval gives the correct average height of any line.

Theorem 44 (Error of the midpoint rule)**.** *If f has a continuous second derivative on $[a, b]$, then there exists a point $\xi \in (a, b)$ such that*

$$\int_a^b f(x)\,dx = S_n + \frac{(b-a)\,h^2}{24}\,f''(\xi).$$

Consequently, if $|f''(x)| \le K$ on $[a, b]$, the error of the midpoint rule is bounded by

$$\left| \int_a^b f(x)\,dx - S_n \right| \le \frac{K}{24}\,\frac{(b-a)^3}{n^4}.$$

Hence the estimate improves rapidly as n grows: doubling n divides the bound by 16.

Meaning

The midpoint rule is the first example of a numerical integration method. It is easy to implement and reasonably accurate for its simplicity, and it illustrates the general idea shared by all these methods: replace the integrand by a simple function, integrate the simple function exactly, and estimate how close the result is to the true integral.

14.7 Generalized (improper) integrals

The Riemann integral was defined for functions that are bounded on a closed and bounded interval $[a, b]$. In applications, however, one often needs to integrate functions that fail one of these assumptions: the interval of integration may be unbounded, or the function may become unbounded somewhere in the interval. In both cases the integral cannot be obtained directly from Cauchy–Riemann sums, but it can be defined by taking a limit of ordinary definite integrals. The resulting objects are called *generalized* (or *improper*) integrals.

Two situations are usually distinguished: integration over an unbounded interval, and integration of an unbounded function. Both are treated with the same idea, a limit over parts of the domain from which the difficulty has been removed. This subsection introduces the general mechanism and then focuses on the second situation, the integration of unbounded functions.

14.7.1 Integration of unbounded functions

The first type of improper integral arises when the function is defined on a half-open interval and becomes unbounded near the excluded endpoint. The integration domain is truncated, the truncated integrals are ordinary (proper) ones, and the limit is then taken.

Definition 41 (Improper integral of an unbounded function)**.** Let f be defined on $[a, b)$ and Riemann integrable on every closed subinterval $[a, t]$ with $a < t < b$, and let f be unbounded as $x \to b^-$. The *generalized integral* of f over $[a, b)$ is defined by the limit

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx,$$

if this limit exists and is finite. In that case the integral is said to *converge* and its value is the limit; otherwise it *diverges*.

If the singularity is the left endpoint a, with f defined on $(a, b]$ and Riemann integrable on every $[t, b]$, the definition is the analogous limit

$$\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx.$$

When the singularity is an interior point $c \in (a, b)$, the domain is split into two half-open intervals at c, and the integral is defined piecewise:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

where each term is the corresponding improper integral. The whole integral converges exactly when both terms converge, and its value is their sum.

The standard comparison example is the integral of the power function near a singularity. For every $\alpha > 0$,

$$\int_0^1 \frac{dx}{x^\alpha}$$

converges if and only if $\alpha < 1$, and in that case its value is $1/(1 - \alpha)$.

A comparison criterion extends the direct comparison test to improper integrals: if $0 \le f(x) \le g(x)$ near the singularity and the generalized integral of g converges, then the generalized integral of f converges as well.

Meaning

Generalized integrals extend the definite integral to the two situations in which it was not defined: unbounded intervals and unbounded functions. The integration of an unbounded function is interpreted as a limit of proper integrals on truncated domains, and convergence means that this limit is finite and independent of how the truncation is performed.

14.7.2 Integrability criteria at a finite point

The convergence of a generalized integral with a finite singularity can often be decided without computing any antiderivative, by comparing the integrand with a function whose integral is known. The reference function is the power $x^{-\alpha}$, whose behaviour near the singularity governs the integral.

Theorem 45 (Convergence of the power integral)**.** *Let $c \in \mathbb{R}$ and $\alpha > 0$. The generalized integral of the power function near the singularity c,*

$$\int_c^{c+\varepsilon} \frac{dx}{|x - c|^\alpha}, \qquad \varepsilon > 0,$$

converges if and only if $\alpha < 1$.

This single result yields two practical criteria.

Theorem 46 (Comparison criterion)**.** *Let f and g be such that the generalized integrals over $[a, b)$ of f and g near the same singular endpoint b are being studied, and let*

$$0 \leq f(x) \leq g(x) \qquad \text{near } b.$$

If the integral of g converges, then the integral of f converges; if the integral of f diverges, then the integral of g diverges.

Theorem 47 (Asymptotic comparison criterion)**.** *Let f and g have constant sign near the singular endpoint b, and assume that the limit*

$$\ell = \lim_{x \to b^-} \frac{f(x)}{g(x)}$$

exists in $(0, \infty)$. Then the generalized integrals of f and g over $[a, b)$ converge or diverge together.

For example, if the integrand behaves near the singular endpoint like a power, the power criterion applies directly:

$$f(x) \sim \frac{C}{(b-x)^\alpha} \quad \Longrightarrow \quad \int_a^b f(x)\,dx \quad \text{converges} \iff \alpha < 1.$$

Meaning

The criteria at a finite point translate the question of convergence into a comparison with a power function. In practice they are the quickest tools for establishing convergence or divergence of an improper integral whose antiderivative is not available, and they prepare the analogous criteria for integrals over an unbounded interval.

14.7.3 Integration over unbounded intervals

The second situation in which the ordinary integral is not defined is that of an unbounded interval of integration. Just as for unbounded functions, the definition proceeds by truncating the domain and passing to a limit.

Definition 42 (Improper integral over an unbounded interval)**.** Let f be Riemann integrable on every closed interval $[a, t]$ with $t \geq a$. The *generalized integral* of f over $[a, \infty)$ is defined by the limit

$$\int_a^\infty f(x)\,dx = \lim_{t \to \infty} \int_a^t f(x)\,dx,$$

if this limit exists and is finite. In that case the integral is said to *converge*, and its value is the limit; otherwise it *diverges*.

For an interval unbounded to the left, the definition is the analogous limit

$$\int_{-\infty}^b f(x)\,dx = \lim_{t \to -\infty} \int_t^b f(x)\,dx,$$

and for the whole real line the interval is split into two parts at any point $c \in \mathbb{R}$:

$$\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^c f(x)\,dx + \int_c^\infty f(x)\,dx,$$

where the integral converges exactly when both terms converge, and its value is their sum.

The reference function for unbounded intervals is again a power. For every $a > 0$ and $\alpha > 0$,

$$\int_a^\infty \frac{dx}{x^\alpha}$$

converges if and only if $\alpha > 1$, and in that case its value is $a^{1-\alpha}/(\alpha - 1)$.

Which of the two behaviours occurs is decided by the comparison criteria, which form the next subsubsection.

14.7.4 Integrability criteria at infinity

The convergence of a generalized integral over an unbounded interval is decided by the same comparison strategy used for finite singularities: the integrand is compared with a power of x, whose integral at infinity is known.

Theorem 48 (Integrability at infinity of the power)**.** *Let $a > 0$. The generalized integral of the power function at infinity,*

$$\int_a^\infty \frac{dx}{x^\alpha}, \qquad \alpha > 0,$$

converges if and only if $\alpha > 1$.

Theorem 49 (Comparison criterion at infinity)**.** *Let f and g be integrable on every closed subinterval of $[a, \infty)$, and let*

$$0 \le f(x) \le g(x) \qquad \text{for } x \text{ large enough.}$$

If $\int_a^\infty g$ converges, then $\int_a^\infty f$ converges; if $\int_a^\infty f$ diverges, then $\int_a^\infty g$ diverges.

Theorem 50 (Asymptotic comparison criterion at infinity)**.** *Let f and g have constant sign for x large, and assume that the limit*

$$\ell = \lim_{x\to\infty} \frac{f(x)}{g(x)}$$

exists in $(0, \infty)$. Then $\int_a^\infty f$ and $\int_a^\infty g$ converge or diverge together.

Combining the asymptotic criterion with the power integral gives the test in its most used form: if the integrand behaves at infinity like a constant over a power of x, the integral converges exactly when the exponent exceeds 1,

$$f(x) \sim \frac{C}{x^\alpha} \quad \Longrightarrow \quad \int_a^\infty f(x)\,dx \quad \text{converges} \iff \alpha > 1.$$

Meaning

Integration over an unbounded interval measures the total amount of a quantity that is spread over an infinite domain. As for finite singularities, the convergence problem is reduced by the comparison criteria to the study of the power function: the same tool works at zero and at infinity, with the roles of the exponent α reversed.

14.8 Integral functions

The fundamental theorem of calculus suggests treating the integral not only as a computational tool but as a way of *defining* functions: a new function can be constructed by integrating a given function with a variable upper limit. A function defined in this way is called an *integral function* (or *accumulation function*).

Definition 43 (Integral function)**.** Let f be Riemann integrable on $[a, b]$ and let $x_0 \in [a, b]$ be a fixed point, called the *initial point*. The *integral function* of f with initial point x_0 is the function $F : [a, b] \to \mathbb{R}$ defined by

$$F(x) = \int_{x_0}^x f(t)\,dt, \qquad x \in [a, b].$$

The variable t is a *dummy variable*: it is bound by the integral sign and does not appear in the result, so the value of F depends only on the upper limit x. The initial point x_0 fixes where the accumulation starts: every integral function satisfies $F(x_0) = 0$, and two integral functions of the same f with initial points x_0 and x_1 differ by the constant $\int_{x_1}^{x_0} f(t)\,dt$, in accordance with the additivity of the integral. If f is merely Riemann integrable, its integral function is well defined and continuous on $[a, b]$; if in addition $|f| \le M$, then $|F(x) - F(y)| \le M\,|x - y|$ for all $x, y \in [a, b]$, so F is even Lipschitz. When f is continuous, the fundamental theorem of calculus, part I, describes the differential behaviour completely.

Theorem 51 (Differential properties of the integral function)**.** *Let f be continuous on $[a,b]$, and let F be the integral function of f with initial point x_0. Then F is differentiable on $[a,b]$ and*

$$F'(x) = f(x), \qquad x \in [a,b].$$

Consequently F is the unique primitive of f that vanishes at x_0. Moreover, F is increasing on every subinterval where $f \geq 0$ and decreasing on every subinterval where $f \leq 0$, and its graph has horizontal tangent at the zeros of f.

The construction produces the elementary functions themselves: for $x > 0$ the natural logarithm is the integral function of $1/t$ with initial point 1,

$$\ln x = \int_1^x \frac{dt}{t},$$

and the arctangent is the integral function of $1/(1+t^2)$ with initial point 0,

$$\arctan x = \int_0^x \frac{dt}{1+t^2}.$$

When no elementary primitive exists, the integral function is the very definition of a new function: the *error function*

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

and the *sine integral*

$$\operatorname{Si}(x) = \int_0^x \frac{\sin t}{t}\, dt$$

have no elementary expression, yet their continuity, differentiability, monotonicity and graphical behaviour follow from the properties of the integral alone.

Meaning

Integral functions shift the viewpoint of the theory: the integral is used to construct functions, not only to compute areas. Every primitive found by the methods of integration is an integral function, and every continuous function generates a primitive in this way, whether or not it can be written in elementary terms.

14.9 The second fundamental theorem of calculus

The *second fundamental theorem of calculus*, also known as *Torricelli's theorem* or the *Newton–Leibniz formula*, evaluates a definite integral by means of any primitive of the integrand, without passing to the limit of Cauchy–Riemann sums.

Theorem 52 (Second fundamental theorem of calculus)**.** *Let f be continuous on $[a,b]$, and let F be any antiderivative of f on $[a,b]$, so that $F' = f$. Then*

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

The difference $F(b) - F(a)$ records the total change of F over $[a,b]$; it is often written with the evaluation notation

$$\Big[F(x)\Big]_a^b = F(b) - F(a).$$

Any primitive may be used: if F_1 and F_2 are both primitives of f on $[a,b]$, then they differ by a constant, and the quantity $F(b) - F(a)$ is the same for both. The two fundamental theorems are converses of one another: the first reconstructs an antiderivative from an integral, the second reconstructs an integral from an antiderivative, and together they show that differentiation and integration are inverse operations.

Meaning

The second fundamental theorem turns the evaluation of every definite integral into the search for a primitive: it is the principle underlying the methods of integration presented in this section. Combined with the first part, it states that the derivative of an integral returns the integrand and that the integral of a derivative measures the total increment of a function.

14.10 Some notions on convolution and linear physical systems

Many physical systems – electrical circuits, thermal devices, mechanical oscillators – are described, at least in first approximation, by *linear* models. A system is *linear* when it satisfies the *superposition principle*: if an input f_1 produces the output y_1 and an input f_2 produces the output y_2, then the input $\alpha f_1 + \beta f_2$ produces the output $\alpha y_1 + \beta y_2$, for all constants α, β. A system is *time-invariant* when a delay of the input produces the same delay of the output. The integral operation that combines the input signal with the response of such a system is the *convolution*.

Definition 44 (Convolution)**.** Let f and g be continuous on $[0, +\infty)$. The *convolution* of f and g is the function

$$(f * g)(x) = \int_0^x f(t)\, g(x - t)\, dt, \qquad x \geq 0.$$

For signals defined on the whole real line the corresponding definition is

$$(f * g)(x) = \int_{-\infty}^{+\infty} f(t)\, g(x - t)\, dt,$$

taken whenever the integral converges.

In the formula the second factor is *reflected* and translated: as t runs from 0 to x, the point $x - t$ runs backwards from x to 0, so $(f * g)(x)$ re-reads the past values of g and weights them with the values of f.

Theorem 53 (Properties of the convolution)**.** *Let f and g be continuous on $[0, +\infty)$. Then $f * g$ exists for every $x \geq 0$ and is continuous on $[0, +\infty)$. The convolution is commutative, $f * g = g * f$; distributive over sums, $f * (g_1 + g_2) = f * g_1 + f * g_2$; homogeneous, $f * (\alpha g) = \alpha\,(f * g)$; and associative, $(f * g) * h = f * (g * h)$. If g is differentiable with continuous derivative, then $f * g$ is differentiable and*

$$(f * g)'(x) = f(x)\, g(0) + \int_0^x f(t)\, g'(x - t)\, dt.$$

In a linear time-invariant system the action of the device on an arbitrary input is completely described by a single function, the *impulse response* h: the output produced by an ideal unit impulse concentrated at time 0 (a limiting notion, made rigorous by the Dirac *delta*, which is not treated here). By time invariance, a pulse of *area* $f(t)\,\Delta t$ concentrated at time t produces, at a later time $x > t$, the response $f(t)\, h(x - t)\,\Delta t$; by superposition, the output at time x is the sum of the responses to all the pulses into which the input history can be decomposed. As $\Delta t \to 0$ the sum becomes an integral:

$$y(x) = \int_0^x f(t)\, h(x - t)\, dt = (f * h)(x),$$

a formula known as *Duhamel's integral*: the output of a linear time-invariant system is the convolution of the input with the impulse response.

For example, the first-order system $\tau\, y'(x) + y(x) = f(x)$, with time constant $\tau > 0$, models both the cooling of a body in a fixed environment and an *RC* circuit; its impulse response is $h(t) = \frac{1}{\tau}\, e^{-t/\tau}$, so that the output is

$$y(x) = \frac{1}{\tau} \int_0^x f(t)\, e^{-(x-t)/\tau}\, dt.$$

The formula exhibits y as a weighted average of the whole past of the input, in which recent instants count more than remote ones, and one verifies that y satisfies $\tau\, y' + y = f$ with $y(0) = 0$.

Meaning

Convolution is the integral counterpart of the superposition principle: the response of a linear system at time x is the integral of all the contributions produced by the past history of the input, each delayed and weighted by the memory of the system. In this way the definite integral appears not only as a computational tool but as the mathematical model of accumulation in physics, and the theory of integration developed in this section becomes the natural language in which linear systems are described.

15 Discrete dynamical models

A *dynamic model* describes the evolution in time of a quantity of interest: the capital of an investment, the size of a population, the price of a good. In the previous sections time was treated as a *continuous* variable, because the functions studied there are defined on intervals of the real line and their behaviour is analyzed with derivatives and integrals. In many applications, however, time is naturally *discrete*: economic quantities are recorded year by year, populations are counted generation after generation, and a bank account is updated at the end of each period. In these situations the state of the system is known only at integer times, and its evolution is a sequence rather than a function of a real variable.

A *discrete dynamical model* describes the evolution of such a quantity through a law that determines the next state from the present one:

$$x_{n+1} = f(x_n), \qquad n = 0, 1, 2, \dots$$

where f is the *evolution law* of the model and the index n plays the role of time. An equation of this form is called a *difference equation* (or *recurrence relation*): it prescribes not the state of the system directly, but the way each state produces the following one.

The evolution is completely determined once the *initial state* x_0 is assigned, because the law f can then be applied repeatedly (*iteration*):

$$x_1 = f(x_0), \qquad x_2 = f(x_1) = f(f(x_0)), \qquad x_3 = f(f(f(x_0))), \ \dots$$

The sequence (x_n) obtained in this way is called the *orbit* (or *trajectory*) of the model starting from x_0. The study of a discrete dynamical model is therefore the study of the iterates of a function, and the theory of sequences developed in §11 – convergence, divergence, monotonicity, recurrence – provides exactly the tools that are needed.

The questions asked of a dynamic model concern the *long-run* behaviour of its orbits:

- does the orbit converge to a finite limit, or does it grow without bound?
- if the limit exists, does it depend on the initial state x_0?
- is the approach to the limit monotone, or does the orbit oscillate?

A special role is played by the states that reproduce themselves: a real number x^* satisfying

$$f(x^*) = x^*$$

is called an *equilibrium* (or *fixed point*) of the model. If the system starts exactly at an equilibrium, it remains there forever, $x_n = x^*$ for every n: the constant sequence is the *stationary solution* of the difference equation, and the convergence of an orbit towards an equilibrium describes the tendency of the model to settle at a stationary regime.

The simplest evolution laws are the *linear* ones,

$$x_{n+1} = q\, x_n + b, \qquad q, b \in \mathbb{R},$$

which already display all the basic phenomena: their orbits may converge to the equilibrium, diverge to infinity, or oscillate, according to the values of q and b. Linear models include the growth of a capital under compound interest, $M_{n+1} = (1+i)\,M_n$, which is a geometric progression in the sense of §2, and the classical supply–demand adjustment of a market. The rest of this section develops these elementary cases first, and then introduces the notions – equilibrium, stability, qualitative behaviour of the orbits – that are needed for general first-order models.

15.1 The Malthus model

The first model we consider describes the growth of a population observed at discrete times, for instance one year or one generation apart. It rests on a single assumption: in every period the population varies by a fixed percentage of its present size. If r denotes this *growth rate*, constant in time, the increment of one period is $r\,P_n$, and the next population is obtained from the present one by adding this increment. The model is meaningful only when the resulting factor is positive, that is $r > -1$: a negative factor would make the population change sign at every step, which has no interpretation.

Definition 45 (Malthus model)**.** Let $r > -1$ be a real constant, called the *growth rate*, and let P_0 be a real number, the *initial population*. The *Malthus model* (or *geometric growth model*) is the discrete dynamical model whose evolution law is

$$P_{n+1} = P_n + r\,P_n = (1+r)\,P_n.$$

It is the linear evolution law of the previous paragraph with $b = 0$ and $q = 1 + r > 0$.

Theorem 54 (Solution of the Malthus model)**.** *For every initial value P_0, the solution of the Malthus model is*

$$P_n = P_0\,(1+r)^n \qquad \textit{for every } n \geq 0.$$

Proof. We argue by induction on n (§7). For $n = 0$ the formula reads $P_0\,(1+r)^0 = P_0$, which is the initial value. If the formula holds for some n, then the equation of the model gives

$$P_{n+1} = (1+r)\,P_n = (1+r)\,P_0\,(1+r)^n = P_0\,(1+r)^{n+1},$$

so it holds for $n + 1$ as well. By induction, the formula holds for every $n \geq 0$. □

The solution is a geometric progression (§2) with initial value P_0 and ratio $q = 1 + r$, and its long-run behaviour follows from the convergence theory of sequences (§11). For a positive initial population $P_0 > 0$:

- if $r > 0$, then $1 + r > 1$ and P_n diverges to $+\infty$: the population grows without bound (exponential growth);
- if $r = 0$, then $P_n = P_0$ for every n: the population is constant, and every initial value is an equilibrium of the model;
- if $-1 < r < 0$, then $0 < 1 + r < 1$ and $P_n \to 0$: the population decays towards zero (exponential decay), without ever becoming negative.

Observe that also the absolute increment $P_{n+1} - P_n = r\,P_n$ is proportional to the current size: the larger the population, the larger the increase from one period to the next.

Doubling time

When $r > 0$ it is useful to know after how many periods the population doubles. Since $P_n/P_0 = (1+r)^n$, the doubling index is the first integer n satisfying

$$(1+r)^n \geq 2 \qquad \Longleftrightarrow \qquad n \geq \frac{\ln 2}{\ln(1+r)},$$

where the equivalence follows from the strict increase of the logarithm. For instance, with a growth rate of 2% per period,

$$\frac{\ln 2}{\ln 1.02} \approx 35 :$$

at two per cent, a population doubles in about 35 periods.

Example

A population of one million individuals grows at the constant rate of 5% per period. Its size after n periods is

$$P_n = 10^6\,(1.05)^n,$$

and it doubles during the 15-th period, since $(1.05)^{14} \approx 1.98$ and $(1.05)^{15} \approx 2.08$, in agreement with the value $\ln 2/\ln 1.05 \approx 14.2$. Replacing the growth rate r with an interest rate i, the same law describes a capital M_n invested at compound interest, $M_{n+1} = (1+i)M_n$: the growth of a capital is a Malthus model with growth rate i.

Meaning

The Malthus model is the mathematical expression of *compound growth*: each period reproduces the present state multiplied by a constant factor, so the population grows by a fixed percentage and, as a function of n, it is the restriction of an exponential function to the integers. Over short horizons, when the rate is genuinely constant, the model describes reality well. Over long horizons its prediction of unbounded exponential growth is unrealistic, because growth is limited by finite resources; this observation, made by Malthus in 1798 together with the remark that subsistence can grow only linearly, motivates the non-linear models, introduced later, in which the growth rate depends on the size of the population.

15.2 The logistic model

The Malthus model assumes that the growth rate is constant in time, so that the population can grow without bound. Over long horizons this is unrealistic: resources are finite, and the larger the population, the harder it is for each individual to find food and space. The simplest correction is to let the growth rate depend on the size of the population, decreasing as the population grows and vanishing when the population reaches the largest size that the environment can sustain, called the *carrying capacity*.

Definition 46 (Logistic model)**.** Let $r > 0$ and $K > 0$ be constants: r is the *unlimited growth rate*, the rate at which the population would grow under Malthusian conditions, and K is the *carrying capacity* of the environment. The *logistic model* is the discrete dynamical model whose evolution law is

$$P_{n+1} = P_n + r\,P_n\Big(1 - \frac{P_n}{K}\Big).$$

The increment of one period is $r\,P_n\big(1 - P_n/K\big)$: it is the Malthusian increment $r\,P_n$ corrected by the factor $1 - P_n/K$, which equals 1 for a vanishing population and decreases to 0 as P_n approaches K. Equivalently, the per-capita growth rate

$$\frac{P_{n+1} - P_n}{P_n} = r\Big(1 - \frac{P_n}{K}\Big)$$

is an affine, decreasing function of the present population: it takes the value r when the population is close to 0 and the value 0 at the carrying capacity. As in the Malthus model, the law is meaningful only when the resulting factor is positive: since $P_{n+1} = P_n\big(1 + r(1 - P_n/K)\big)$, this requires $P_n < K\,(1+r)/r$; a population beyond this bound would change sign at the next period, which has no interpretation.

Equilibria

Theorem 55 (Equilibria of the logistic model)**.** *The equilibria of the logistic model are $P^* = 0$ and $P^* = K$.*

Proof. An equilibrium is a solution of $P^* = f(P^*)$, where $f(P) = P + r\,P\big(1 - P/K\big)$ is the evolution law. Substituting the law gives

$$P^* = P^* + r\,P^*\Big(1 - \frac{P^*}{K}\Big) \qquad \Longleftrightarrow \qquad r\,P^*\Big(1 - \frac{P^*}{K}\Big) = 0,$$

and since $r > 0$ the last equation holds exactly when $P^* = 0$ or $P^* = K$. □

The two equilibria are the stationary solutions of the model: a population that starts at 0 remains extinct, and a population that starts exactly at the carrying capacity remains there for every n. The equilibrium $P^* = K$ describes a population saturated at the maximum size that the environment can support; the analysis of the orbits shows that it is the one towards which the model tends.

Qualitative behaviour

Unlike the Malthus model, the logistic model admits no closed-form solution: P_n is obtained by iterating a quadratic function, and no simple formula expresses P_n in terms of n and P_0. The study of the model is therefore qualitative, and it rests on the sign of the increment $P_{n+1} - P_n = r\,P_n\big(1 - P_n/K\big)$. For a positive initial value:

- if $0 < P_n < K$, the increment is positive: the population increases;
- if $P_n = K$, the increment is zero: the population stays at the equilibrium;
- if $P_n > K$, the increment is negative: the population decreases.

In either case the next population lies on the side of K towards which the increment points: below the carrying capacity the population grows, above it the population declines. Moreover, the increment is largest when $P_n = K/2$: as a function of P_n, the quantity $r\,P_n(1 - P_n/K)$ is a second-degree function with negative leading coefficient, whose maximum $rK/4$ is attained at $P_n = K/2$. The population grows fastest, in absolute terms, when it is half of the carrying capacity.

For small values of the growth rate the approach to the equilibrium is monotone.

Theorem 56 (Convergence to the carrying capacity)**.** *If $0 < r \le 1$ and $0 < P_0 \le K$, then the orbit of the logistic model is non-decreasing and converges to K.*

Proof. If $P_0 = K$ the orbit is constant, so assume $0 < P_0 < K$. We prove by induction that $0 < P_n < K$ for every n and that the sequence is strictly increasing. Suppose $0 < P_n < K$ and write $x_n = P_n/K \in (0,1)$. The increment is positive,

$$P_{n+1} - P_n = r\,P_n\,(1 - x_n) > 0,$$

so $P_{n+1} > P_n > 0$. For the upper bound, a direct computation gives

$$P_{n+1} - K = K\,(1 - x_n)(r\,x_n - 1) < 0,$$

because $1 - x_n > 0$ and, since $r \le 1$, also $r\,x_n < 1$. Hence $P_n < P_{n+1} < K$, which completes the induction: the sequence (P_n) is strictly increasing and bounded above by K, so it converges to a limit L with $P_0 \le L \le K$; in particular $L > 0$. The evolution law has the form $P_{n+1} = f(P_n)$ with $f(P) = P + r\,P(1 - P/K)$, a polynomial, which is continuous (§11); therefore $f(P_n) \to f(L)$, while $f(P_n) = P_{n+1} \to L$ as well, so $L = f(L)$, that is

$$r\,L\Big(1 - \frac{L}{K}\Big) = 0.$$

Since $r > 0$ and $L > 0$, the only possibility is $L = K$. □

For larger values of the rate the orbits may overshoot the carrying capacity and approach it by oscillations. The mechanism is visible if one subtracts K from both sides of the evolution law: writing $e_n = P_n - K$ for the error with respect to the equilibrium, a direct computation gives the exact identity

$$e_{n+1} = (1 - r)\,e_n - \frac{r}{K}\,e_n^2.$$

When the orbit is close to K the quadratic term is small, and the error is multiplied at each period by a factor close to $1 - r$. Hence:

- if $0 < r < 1$, then $0 < 1 - r < 1$: the error keeps its sign and decreases in magnitude, and the orbit converges to K monotonically, as in the theorem;
- if $1 < r < 2$, then $-1 < 1 - r < 0$: the error changes sign at every period and decreases in magnitude, and the orbit approaches K by damped oscillations; it can be proved that the orbit still converges to K for every initial value $0 < P_0 < K(1 + r)/r$, the range in which the model is meaningful;
- if $r > 2$, then $|1 - r| > 1$: small errors are amplified at every period, the equilibrium K is unstable, and a typical orbit does not approach it.

For $r > 2$ the quadratic correction prevents the orbit from escaping to infinity and new phenomena appear: for a first range of values of r the orbit settles on a periodic cycle of period 2; as r increases, cycles of period 4, 8, 16, … succeed one another, and beyond a limit value of r the behaviour becomes aperiodic and extremely sensitive to the initial value, a phenomenon known as *chaos*. The logistic model is the simplest model in which this transition can be observed.

Example

Let $K = 1000$ and $P_0 = 100$. With the moderate rate $r = 0.5$ the first iterates, rounded to integers, are

$$P_1 = 145, \quad P_2 = 207, \quad P_3 = 289, \quad P_4 = 392, \quad P_5 = 511, \quad P_6 = 636,$$

and the sequence increases towards the carrying capacity, in agreement with the theorem. With the larger rate $r = 1.9$ the first iterates are

$$P_1 = 271, \quad P_2 = 646, \quad P_3 = 1081, \quad P_4 = 915, \quad P_5 = 1063, \quad P_6 = 936,$$

the orbit overshoots the carrying capacity and then oscillates around it with decreasing amplitude: this is the damped oscillation described above.

Meaning

The logistic model is the mathematical expression of *constrained growth*: each period reproduces the Malthusian increment corrected by the factor $1 - P_n/K$, which measures how close the population is to saturation. The long-run prediction is no longer unbounded exponential growth, as in the Malthus model, but convergence towards the carrying capacity, in agreement with the observation that real populations tend to stabilize. The model was introduced by Verhulst in 1838 and is used today in ecology, in demography and in economics. Its main theoretical interest is that a single, fixed evolution law produces qualitatively different long-run behaviours as the parameter r varies – monotone convergence, damped oscillations, periodic cycles, chaos – and this makes it the prototype of non-linear discrete dynamical models.